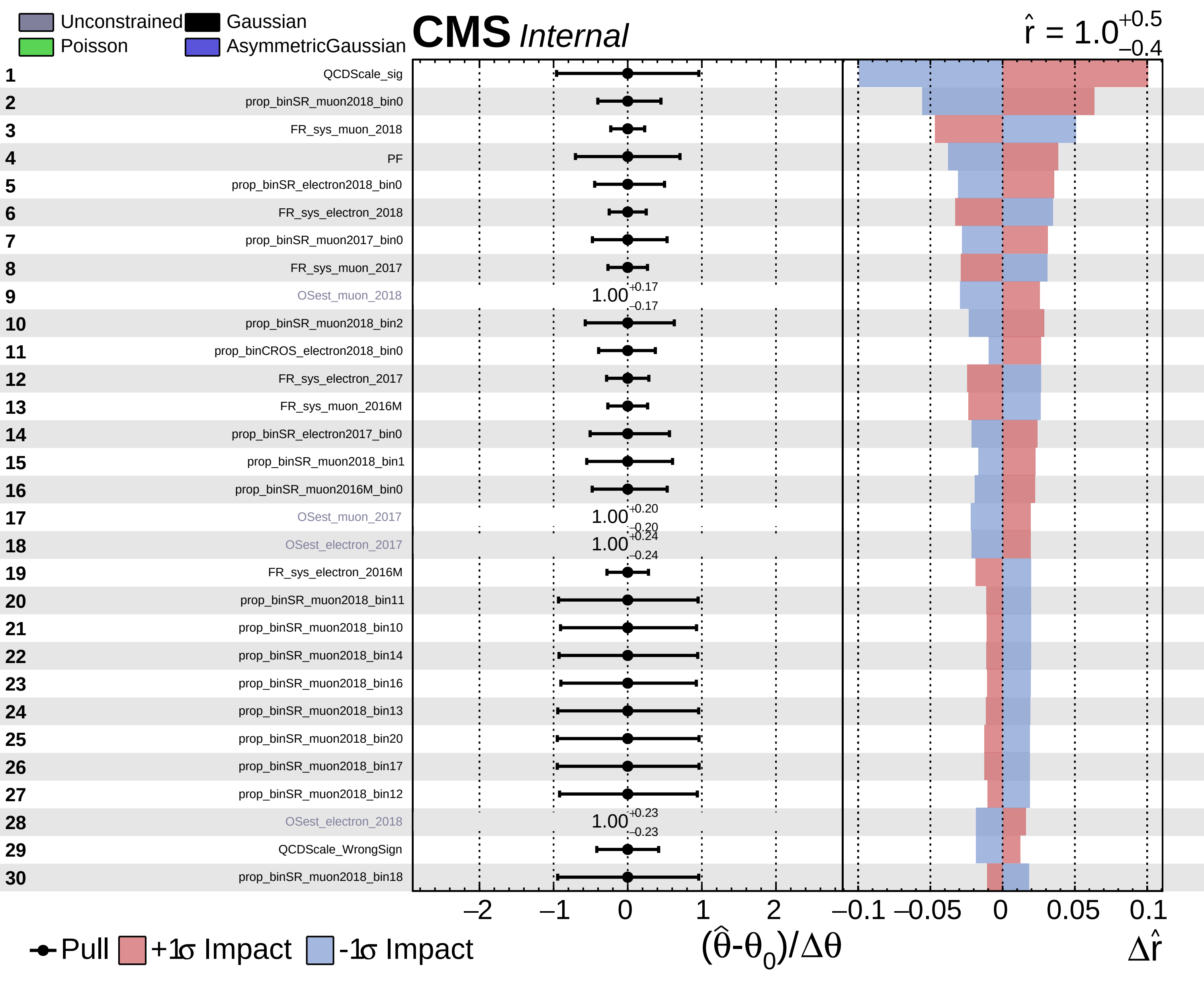
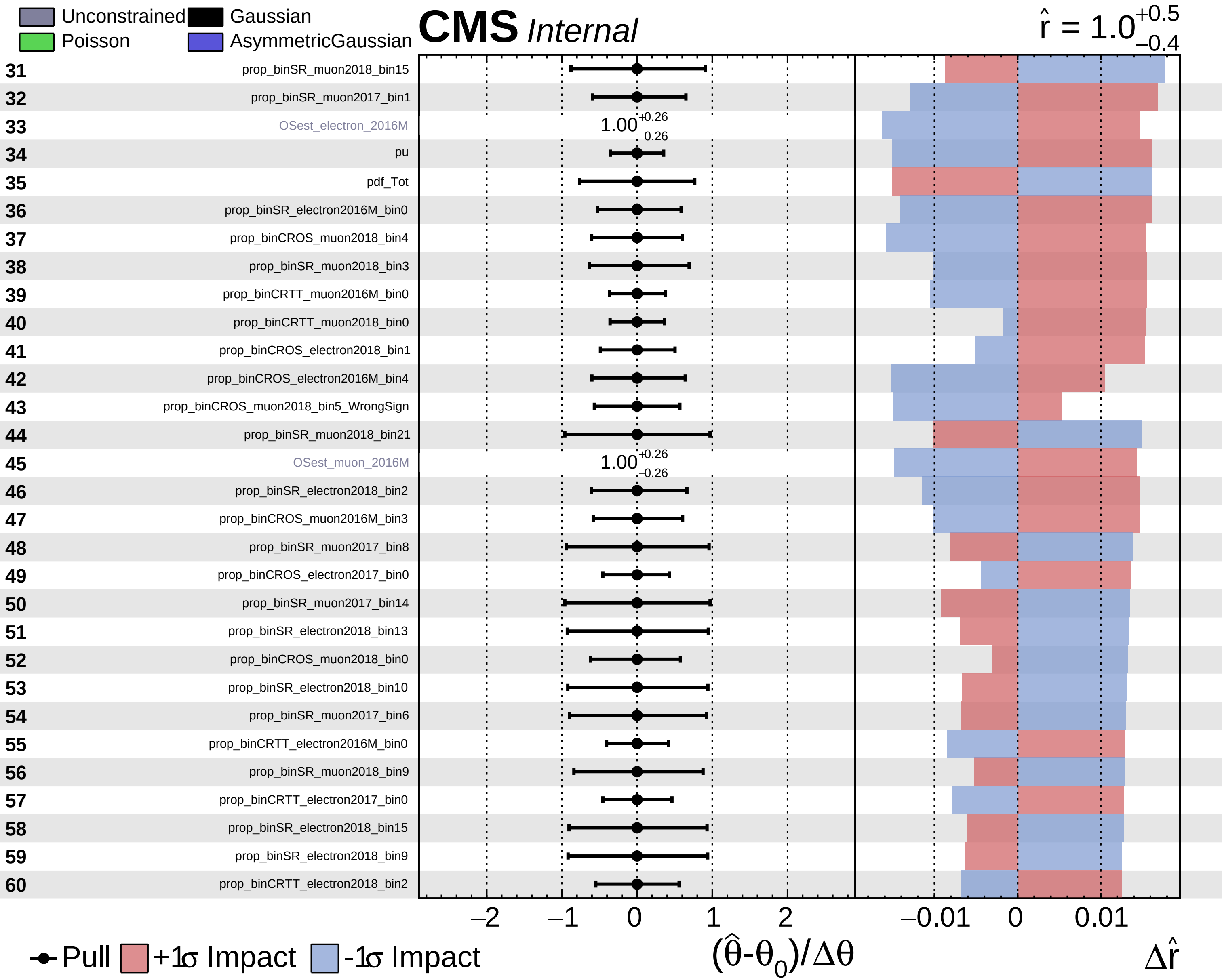
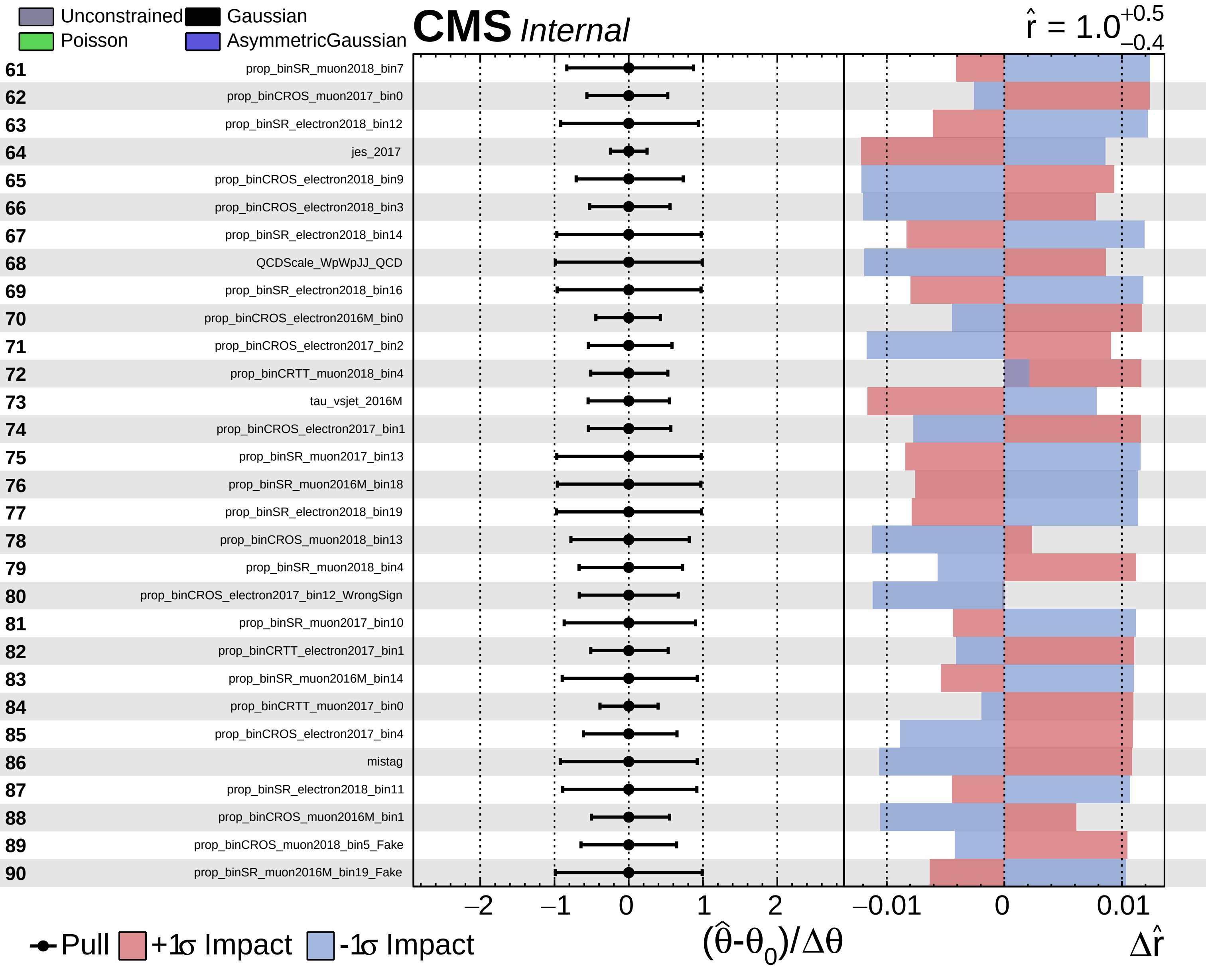


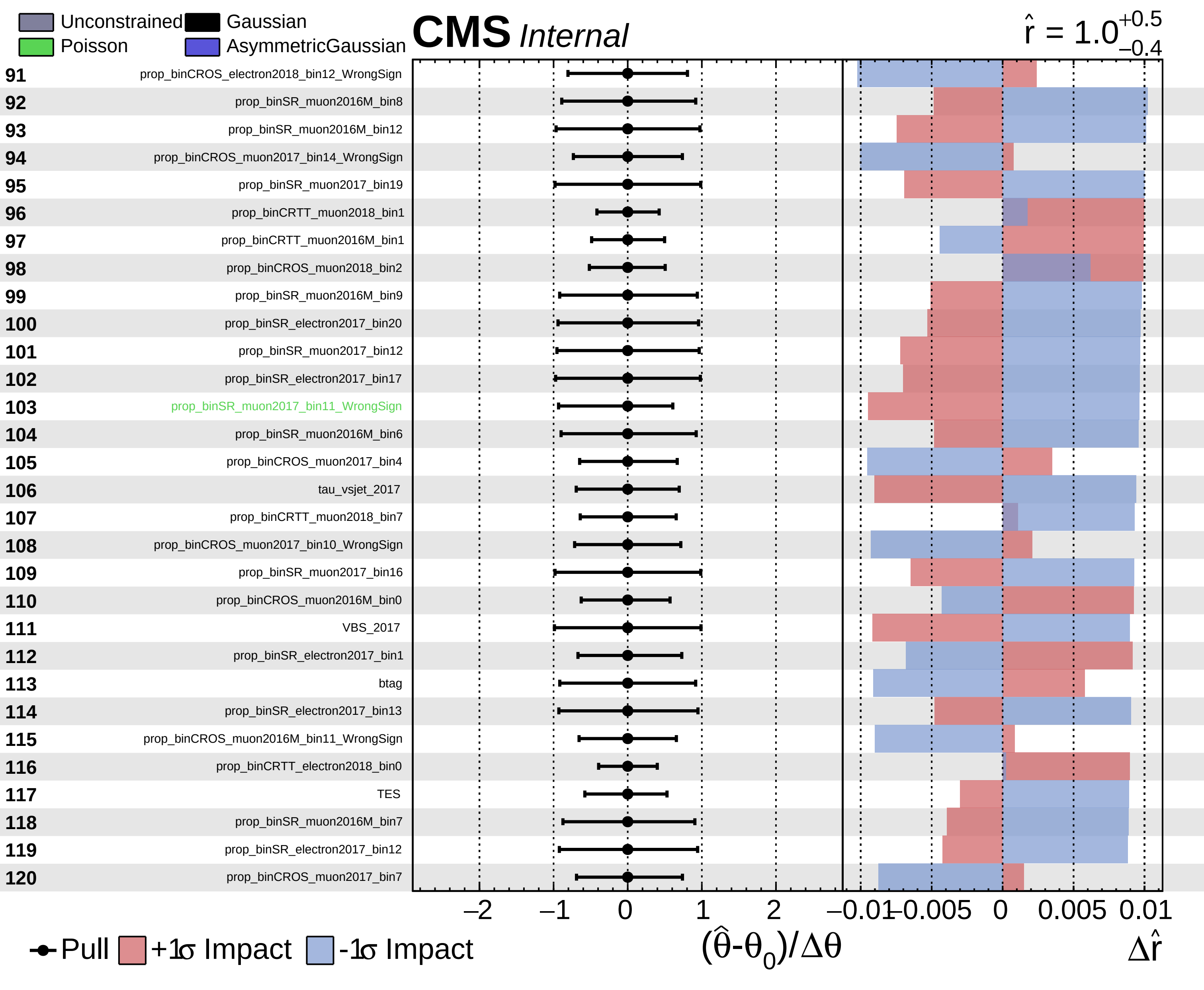
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
1 QCDScale_sig
2 prop_binSR_muon2018_bin0
3 FR_sys_muon_2018
4 PF
5 prop_binSR_electron2018_bin0
6 FR_sys_electron_2018
7 prop_binSR_muon2017_bin0
8 FR_sys_muon_2017
9 OSest_muon_2018
$1.00^{+0.17}_{-0.17}$
10 prop_binSR_muon2018_bin2
11 prop_binCROS_electron2018_bin0
12 FR_sys_electron_2017
13 FR_sys_muon_2016M
14 prop_binSR_electron2017_bin0
15 prop_binSR_muon2018_bin1
16 prop_binSR_muon2016M_bin0
17 OSest_muon_2017
$1.00^{+0.20}_{-0.20}$
18 OSest_electron_2017
$1.00^{+0.24}_{-0.24}$
19 FR_sys_electron_2016M
20 prop_binSR_muon2018_bin11
21 prop_binSR_muon2018_bin10
22 prop_binSR_muon2018_bin14
23 prop_binSR_muon2018_bin16
24 prop_binSR_muon2018_bin13
25 prop_binSR_muon2018_bin20
26 prop_binSR_muon2018_bin17
27 prop_binSR_muon2018_bin12
28 OSest_electron_2018
$1.00^{+0.23}_{-0.23}$
29 QCDScale_WrongSign
30 prop_binSR_muon2018_bin18
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



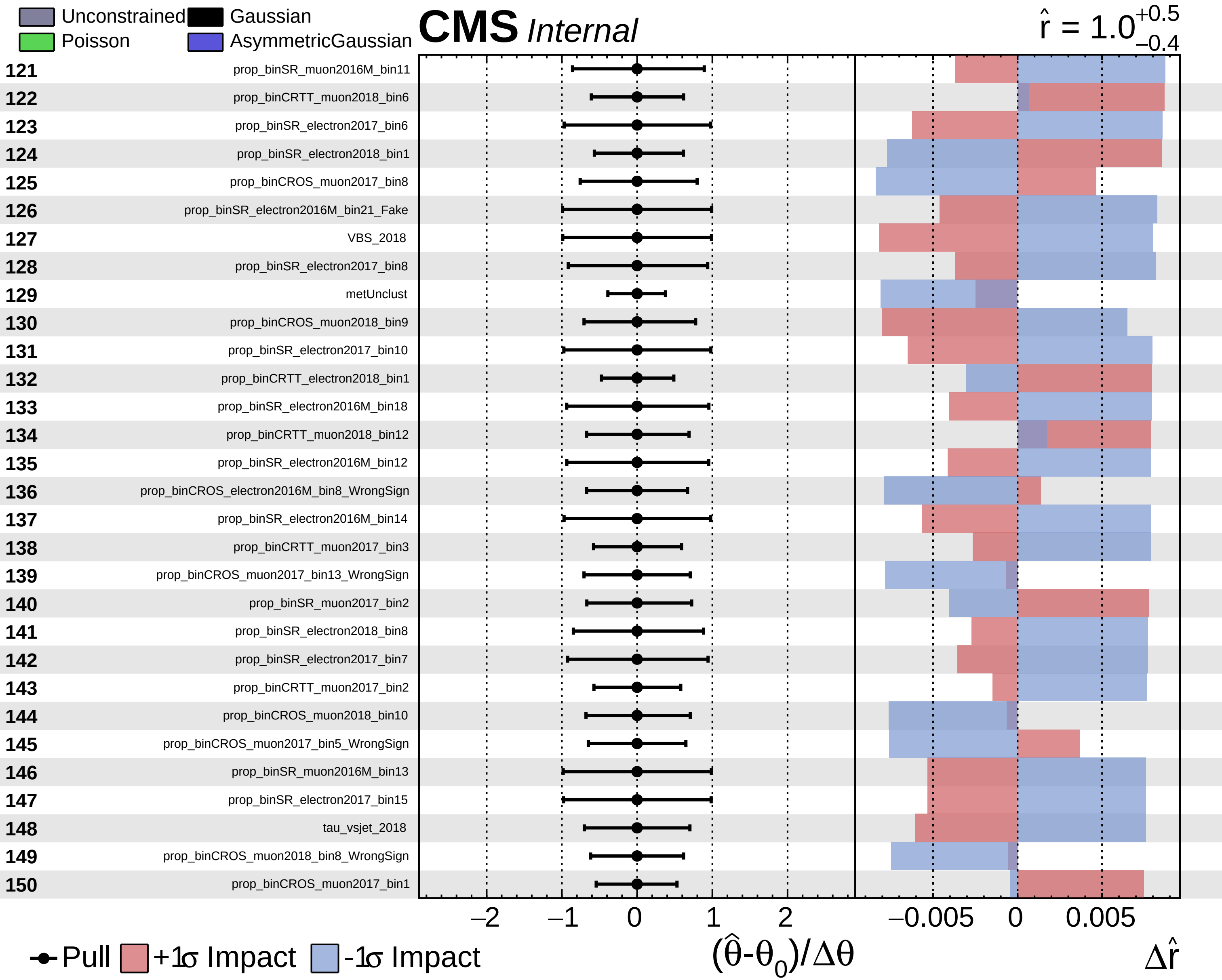
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0+0.5 −0.4
31 prop_binSR_muon2018_bin15
32 prop_binSR_muon2017_bin1
33 OSest_electron_2016M 1.00 +0.26 −0.26
34 pu
35 pdf_Tot
36 prop_binSR_electron2016M_bin0
37 prop_binCROS_muon2018_bin4
38 prop_binSR_muon2018_bin3
39 prop_binCRTT_muon2016M_bin0
40 prop_binCRTT_muon2018_bin0
41 prop_binCROS_electron2018_bin1
42 prop_binCROS_electron2016M_bin4
43 prop_binCROS_muon2018_bin5_WrongSign
44 prop_binSR_muon2018_bin21
45 OSest_muon_2016M 1.00 +0.26 −0.26
46 prop_binSR_electron2018_bin2
47 prop_binCROS_muon2016M_bin3
48 prop_binSR_muon2017_bin8
49 prop_binCROS_electron2017_bin0
50 prop_binSR_muon2017_bin14
51 prop_binSR_electron2018_bin13
52 prop_binCROS_muon2018_bin0
53 prop_binSR_electron2018_bin10
54 prop_binSR_muon2017_bin6
55 prop_binCRTT_electron2016M_bin0
56 prop_binSR_muon2018_bin9
57 prop_binCRTT_electron2017_bin0
58 prop_binSR_electron2018_bin15
59 prop_binSR_electron2018_bin9
60 prop_binCRTT_electron2018_bin2
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



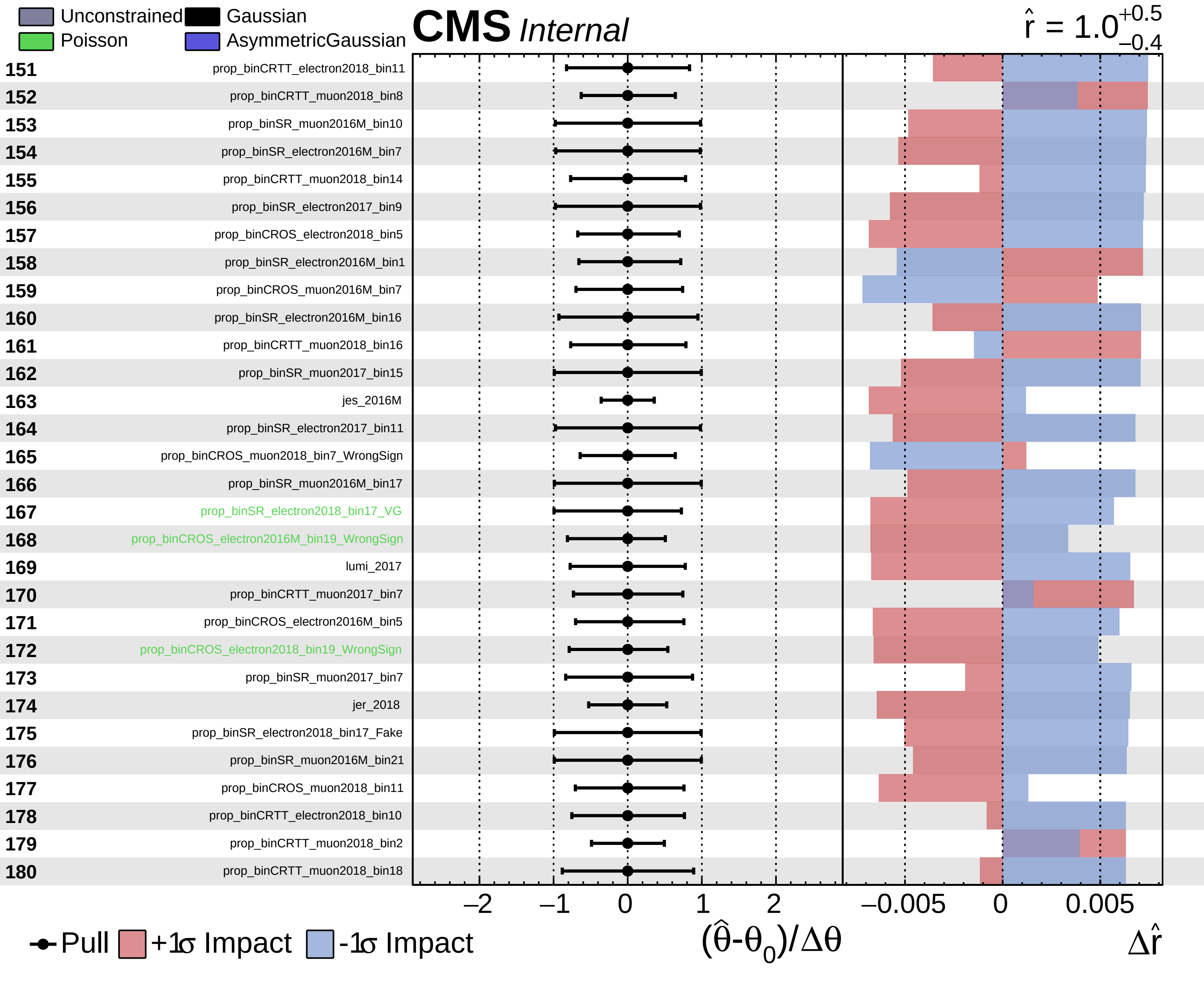
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
61 prop_binSR_muon2018_bin7
62 prop_binCROS_muon2017_bin0
63 prop_binSR_electron2018_bin12
64 jes_2017
65 prop_binCROS_electron2018_bin9
66 prop_binCROS_electron2018_bin3
67 prop_binSR_electron2018_bin14
68 QCDScale_WpWpJJ_QCD
69 prop_binSR_electron2018_bin16
70 prop_binCROS_electron2016M_bin0
71 prop_binCROS_electron2017_bin2
72 prop_binCRTT_muon2018_bin4
73 tau_vsjet_2016M
74 prop_binCROS_electron2017_bin1
75 prop_binSR_muon2017_bin13
76 prop_binSR_muon2016M_bin18
77 prop_binSR_electron2018_bin19
78 prop_binCROS_muon2018_bin13
79 prop_binSR_muon2018_bin4
80 prop_binCROS_electron2017_bin12_WrongSign
81 prop_binSR_muon2017_bin10
82 prop_binCRTT_electron2017_bin1
83 prop_binSR_muon2016M_bin14
84 prop_binCRTT_muon2017_bin0
85 prop_binCROS_electron2017_bin4
86 mistag
87 prop_binSR_electron2018_bin11
88 prop_binCROS_muon2016M_bin1
89 prop_binCROS_muon2018_bin5_Fake
90 prop_binSR_muon2016M_bin19_Fake
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

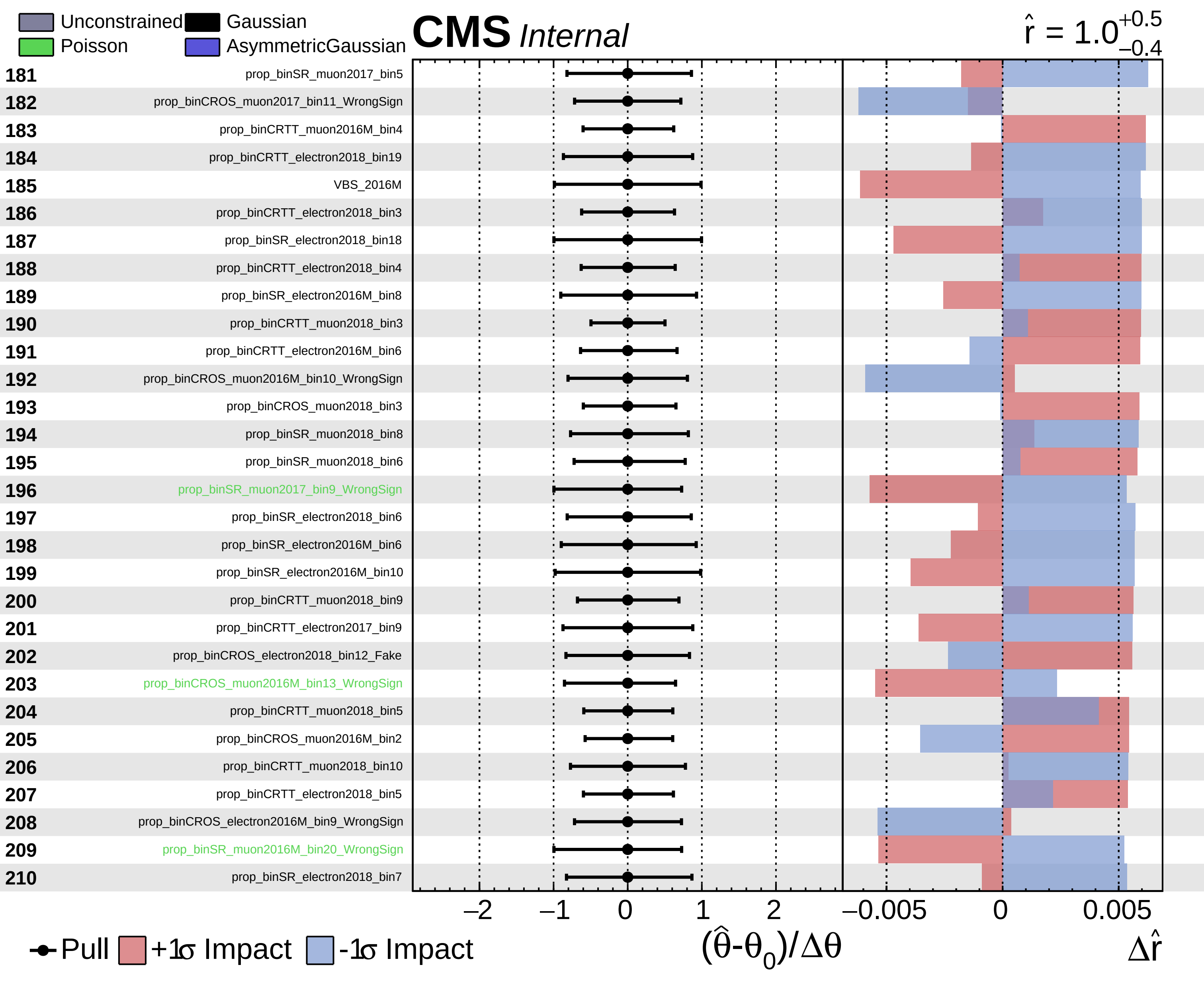


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
91 prop_binCROS_electron2018_bin12_WrongSign
92 prop_binSR_muon2016M_bin8
93 prop_binSR_muon2016M_bin12
94 prop_binCROS_muon2017_bin14_WrongSign
95 prop_binSR_muon2017_bin19
96 prop_binCRTT_muon2018_bin1
97 prop_binCRTT_muon2016M_bin1
98 prop_binCROS_muon2018_bin2
99 prop_binSR_muon2016M_bin9
100 prop_binSR_electron2017_bin20
101 prop_binSR_muon2017_bin12
102 prop_binSR_electron2017_bin17
103 prop_binSR_muon2017_bin11_WrongSign
104 prop_binSR_muon2016M_bin6
105 prop_binCROS_muon2017_bin4
106 tau_vsjet_2017
107 prop_binCRTT_muon2018_bin7
108 prop_binCROS_muon2017_bin10_WrongSign
109 prop_binSR_muon2017_bin16
110 prop_binCROS_muon2016M_bin0
111 VBS_2017
112 prop_binSR_electron2017_bin1
113 btag
114 prop_binSR_electron2017_bin13
115 prop_binCROS_muon2016M_bin11_WrongSign
116 prop_binCRTT_electron2018_bin0
117 TES
118 prop_binSR_muon2016M_bin7
119 prop_binSR_electron2017_bin12
120 prop_binCROS_muon2017_bin7
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

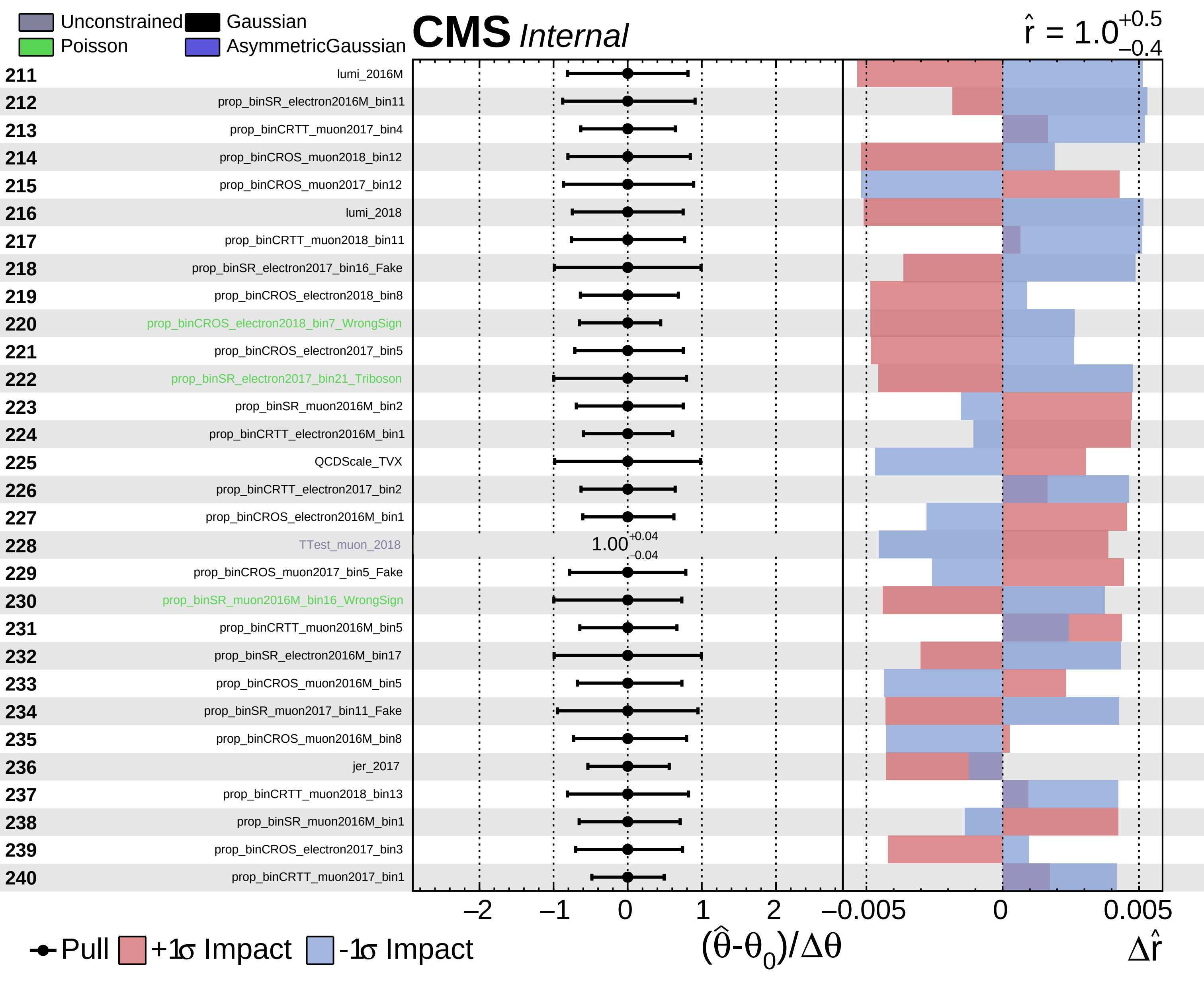


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
121 prop_binSR_muon2016M_bin11
122 prop_binCRTT_muon2018_bin6
123 prop_binSR_electron2017_bin6
124 prop_binSR_electron2018_bin1
125 prop_binCROS_muon2017_bin8
126 prop_binSR_electron2016M_bin21_Fake
127 VBS_2018
128 prop_binSR_electron2017_bin8
129 metUnclust
130 prop_binCROS_muon2018_bin9
131 prop_binSR_electron2017_bin10
132 prop_binCRTT_electron2018_bin1
133 prop_binSR_electron2016M_bin18
134 prop_binCRTT_muon2018_bin12
135 prop_binSR_electron2016M_bin12
136 prop_binCROS_electron2016M_bin8_WrongSign
137 prop_binSR_electron2016M_bin14
138 prop_binCRTT_muon2017_bin3
139 prop_binCROS_muon2017_bin13_WrongSign
140 prop_binSR_muon2017_bin2
141 prop_binSR_electron2018_bin8
142 prop_binSR_electron2017_bin7
143 prop_binCRTT_muon2017_bin2
144 prop_binCROS_muon2018_bin10
145 prop_binCROS_muon2017_bin5_WrongSign
146 prop_binSR_muon2016M_bin13
147 prop_binSR_electron2017_bin15
148 tau_vsjet_2018
149 prop_binCROS_muon2018_bin8_WrongSign
150 prop_binCROS_muon2017_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

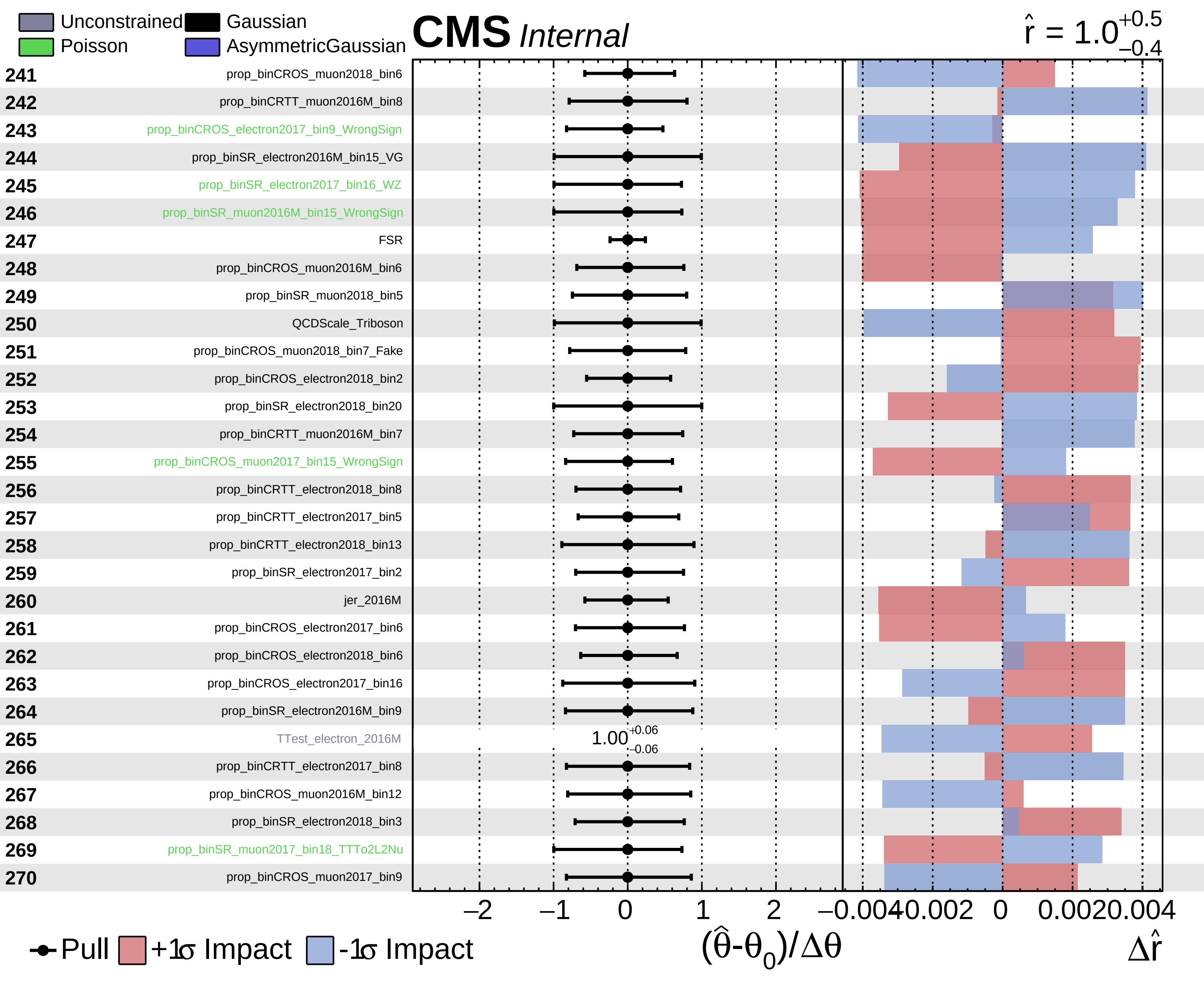

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
151 prop_binCRTT_electron2018_bin11
152 prop_binCRTT_muon2018_bin8
153 prop_binSR_muon2016M_bin10
154 prop_binSR_electron2016M_bin7
155 prop_binCRTT_muon2018_bin14
156 prop_binSR_electron2017_bin9
157 prop_binCROS_electron2018_bin5
158 prop_binSR_electron2016M_bin1
159 prop_binCROS_muon2016M_bin7
160 prop_binSR_electron2016M_bin16
161 prop_binCRTT_muon2018_bin16
162 prop_binSR_muon2017_bin15
163 jes_2016M
164 prop_binSR_electron2017_bin11
165 prop_binCROS_muon2018_bin7_WrongSign
166 prop_binSR_muon2016M_bin17
167 prop_binSR_electron2018_bin17_VG
168 prop_binCROS_electron2016M_bin19_WrongSign
169 lumi_2017
170 prop_binCRTT_muon2017_bin7
171 prop_binCROS_electron2016M_bin5
172 prop_binCROS_electron2018_bin19_WrongSign
173 prop_binSR_muon2017_bin7
174 jer_2018
175 prop_binSR_electron2018_bin17_Fake
176 prop_binSR_muon2016M_bin21
177 prop_binCROS_muon2018_bin11
178 prop_binCRTT_electron2018_bin10
179 prop_binCRTT_muon2018_bin2
180 prop_binCRTT_muon2018_bin18
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

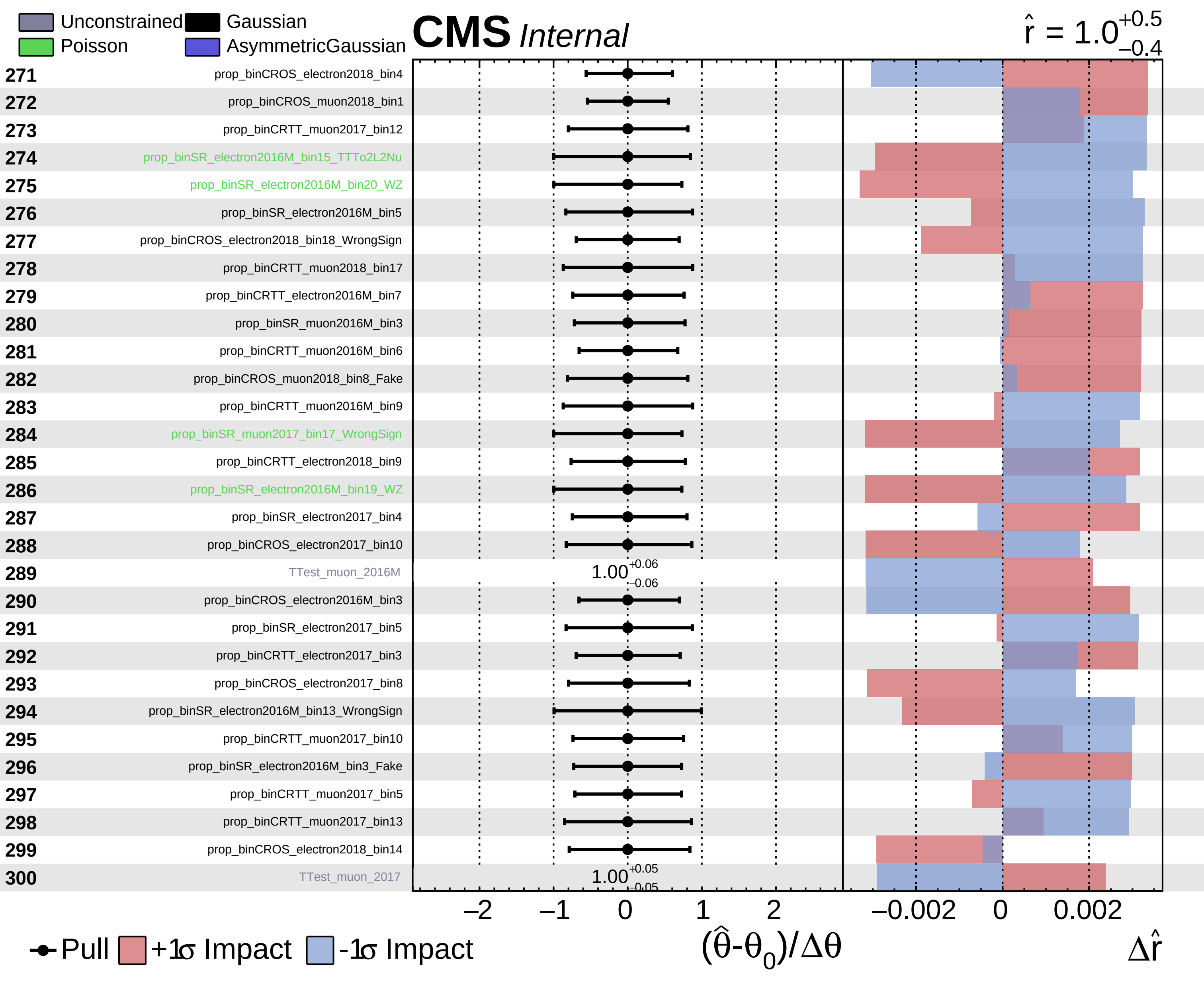


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
181 prop_binSR_muon2017_bin5
182 prop_binCROS_muon2017_bin11_WrongSign
183 prop_binCRTT_muon2016M_bin4
184 prop_binCRTT_electron2018_bin19
185 VBS_2016M
186 prop_binCRTT_electron2018_bin3
187 prop_binSR_electron2018_bin18
188 prop_binCRTT_electron2018_bin4
189 prop_binSR_electron2016M_bin8
190 prop_binCRTT_muon2018_bin3
191 prop_binCRTT_electron2016M_bin6
192 prop_binCROS_muon2016M_bin10_WrongSign
193 prop_binCROS_muon2018_bin3
194 prop_binSR_muon2018_bin8
195 prop_binSR_muon2018_bin6
196 prop_binSR_muon2017_bin9_WrongSign
197 prop_binSR_electron2018_bin6
198 prop_binSR_electron2016M_bin6
199 prop_binSR_electron2016M_bin10
200 prop_binCRTT_muon2018_bin9
201 prop_binCRTT_electron2017_bin9
202 prop_binCROS_electron2018_bin12_Fake
203 prop_binCROS_muon2016M_bin13_WrongSign
204 prop_binCRTT_muon2018_bin5
205 prop_binCROS_muon2016M_bin2
206 prop_binCRTT_muon2018_bin10
207 prop_binCRTT_electron2018_bin5
208 prop_binCROS_electron2016M_bin9_WrongSign
209 prop_binSR_muon2016M_bin20_WrongSign
210 prop_binSR_electron2018_bin7
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

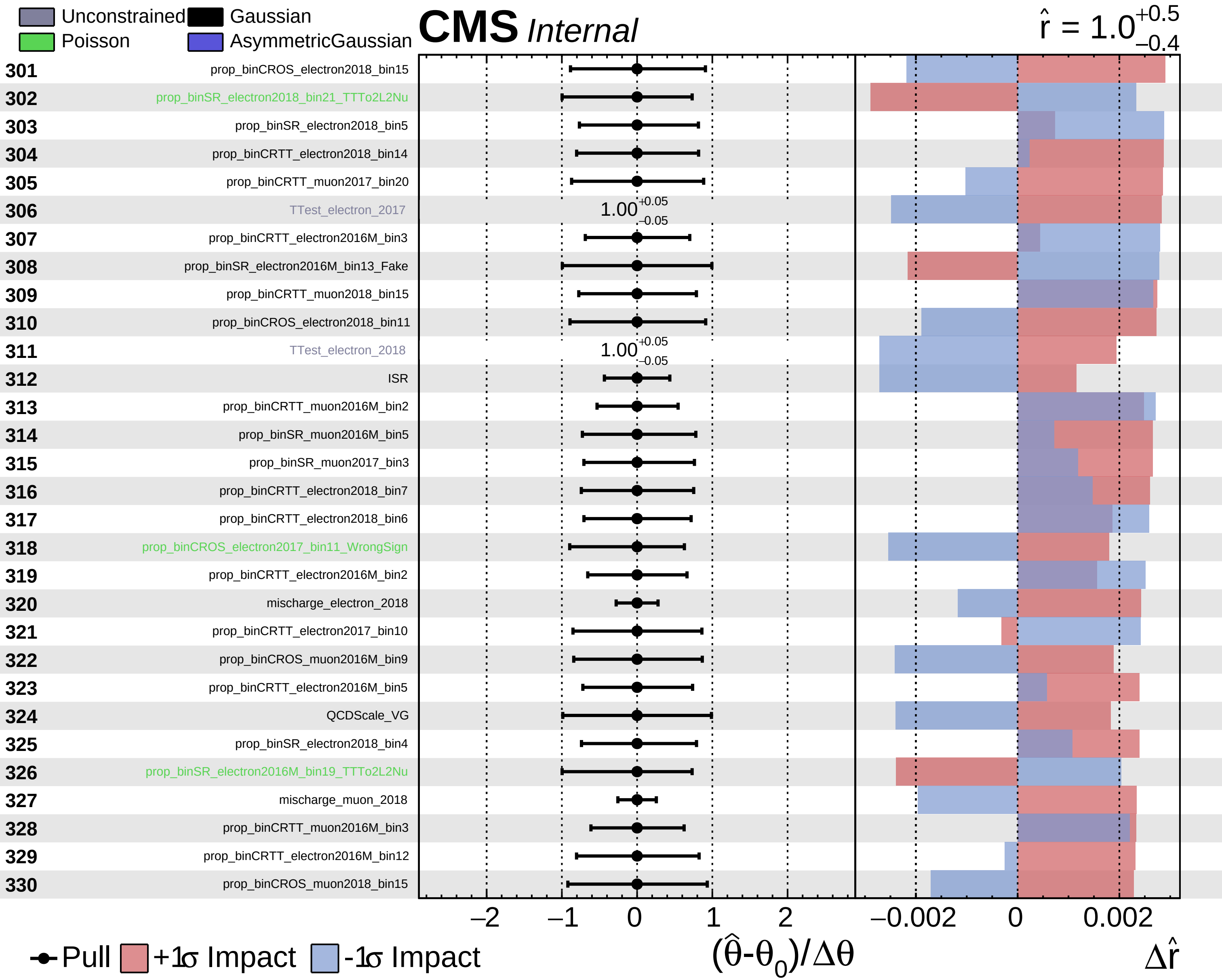


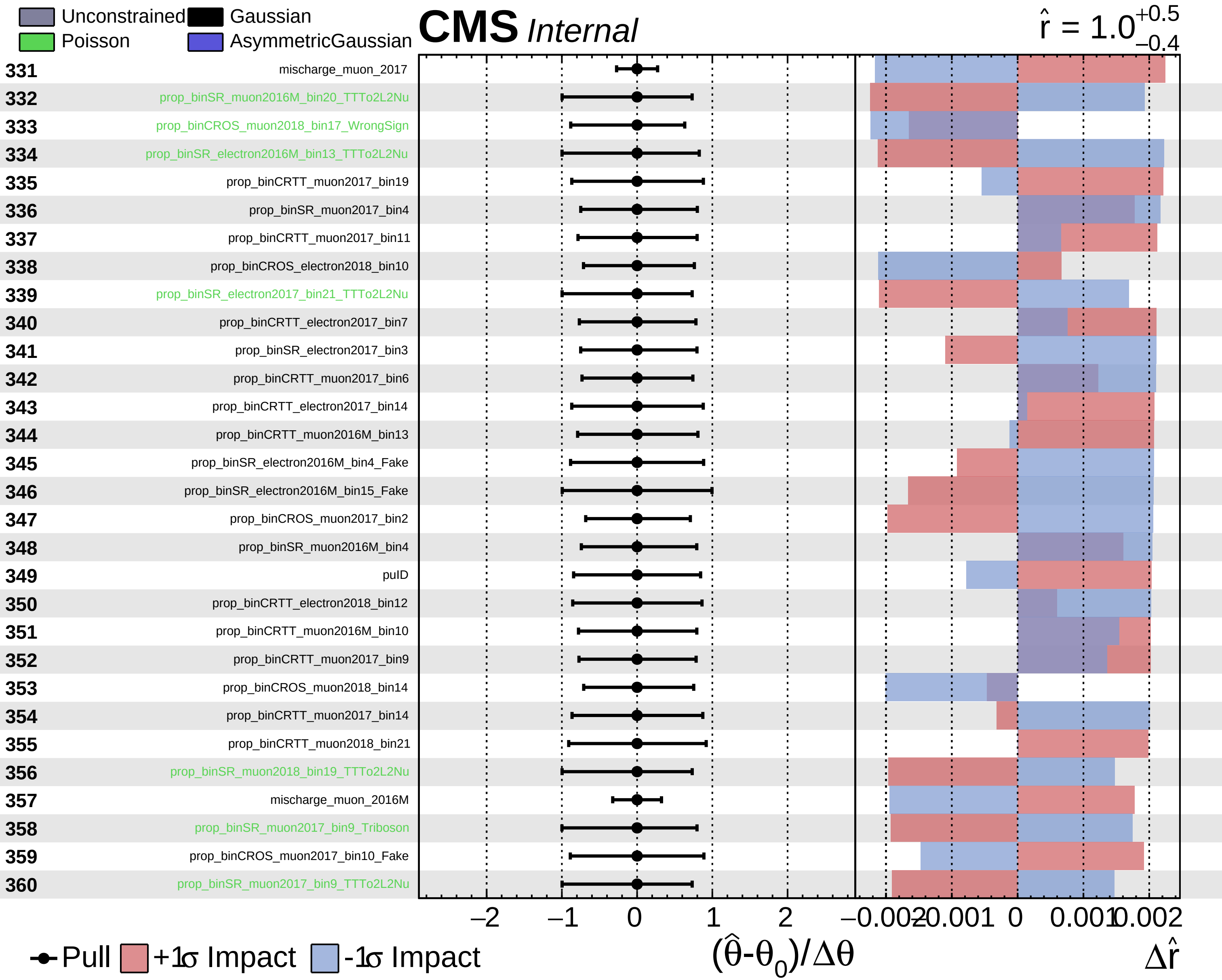
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
211 lumi_2016M
212 prop_binSR_electron2016M_bin11
213 prop_binCRTT_muon2017_bin4
214 prop_binCROS_muon2018_bin12
215 prop_binCROS_muon2017_bin12
216 lumi_2018
217 prop_binCRTT_muon2018_bin11
218 prop_binSR_electron2017_bin16_Fake
219 prop_binCROS_electron2018_bin8
220 prop_binCROS_electron2018_bin7_WrongSign
221 prop_binCROS_electron2017_bin5
222 prop_binSR_electron2017_bin21_Triboson
223 prop_binSR_muon2016M_bin2
224 prop_binCRTT_electron2016M_bin1
225 QCDScale_TVX
226 prop_binCRTT_electron2017_bin2
227 prop_binCROS_electron2016M_bin1
228 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
229 prop_binCROS_muon2017_bin5_Fake
230 prop_binSR_muon2016M_bin16_WrongSign
231 prop_binCRTT_muon2016M_bin5
232 prop_binSR_electron2016M_bin17
233 prop_binCROS_muon2016M_bin5
234 prop_binSR_muon2017_bin11_Fake
235 prop_binCROS_muon2016M_bin8
236 jer_2017
237 prop_binCRTT_muon2018_bin13
238 prop_binSR_muon2016M_bin1
239 prop_binCROS_electron2017_bin3
240 prop_binCRTT_muon2017_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
241 prop_binCROS_muon2018_bin6
242 prop_binCRTT_muon2016M_bin8
243 prop_binCROS_electron2017_bin9_WrongSign
244 prop_binSR_electron2016M_bin15_VG
245 prop_binSR_electron2017_bin16_WZ
246 prop_binSR_muon2016M_bin15_WrongSign
247 FSR
248 prop_binCROS_muon2016M_bin6
249 prop_binSR_muon2018_bin5
250 QCDScale_Triboson
251 prop_binCROS_muon2018_bin7_Fake
252 prop_binCROS_electron2018_bin2
253 prop_binSR_electron2018_bin20
254 prop_binCRTT_muon2016M_bin7
255 prop_binCROS_muon2017_bin15_WrongSign
256 prop_binCRTT_electron2018_bin8
257 prop_binCRTT_electron2017_bin5
258 prop_binCRTT_electron2018_bin13
259 prop_binSR_electron2017_bin2
260 jer_2016M
261 prop_binCROS_electron2017_bin6
262 prop_binCROS_electron2018_bin6
263 prop_binCROS_electron2017_bin16
264 prop_binSR_electron2016M_bin9
265 TTest_electron_2016M
1.00 +0.06 −0.06
266 prop_binCRTT_electron2017_bin8
267 prop_binCROS_muon2016M_bin12
268 prop_binSR_electron2018_bin3
269 prop_binSR_muon2017_bin18_TTTo2L2Nu
270 prop_binCROS_muon2017_bin9
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

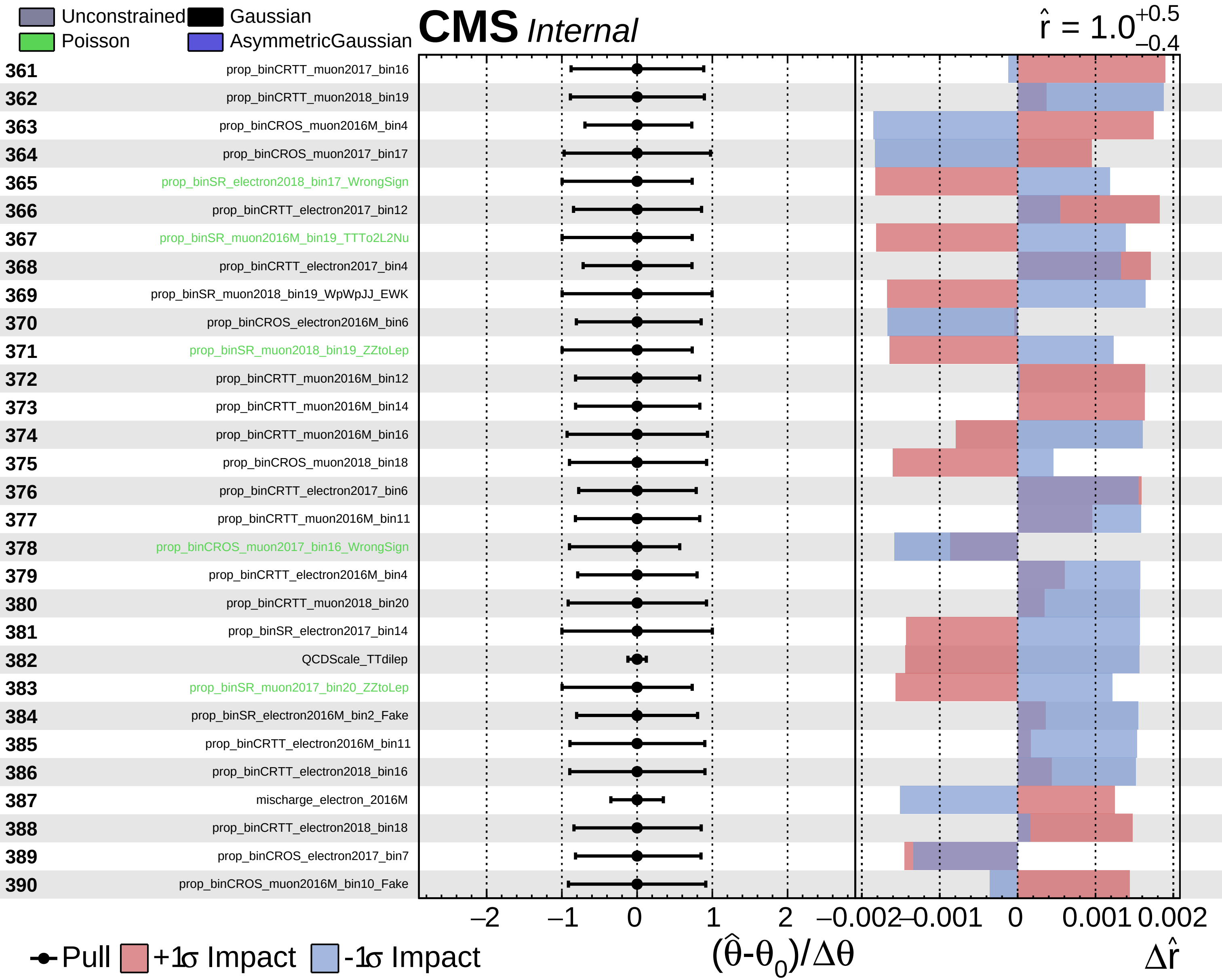


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
271 prop_binCROS_electron2018_bin4
272 prop_binCROS_muon2018_bin1
273 prop_binCRTT_muon2017_bin12
274 prop_binSR_electron2016M_bin15_TTTo2L2Nu
275 prop_binSR_electron2016M_bin20_WZ
276 prop_binSR_electron2016M_bin5
277 prop_binCROS_electron2018_bin18_WrongSign
278 prop_binCRTT_muon2018_bin17
279 prop_binCRTT_electron2016M_bin7
280 prop_binSR_muon2016M_bin3
281 prop_binCRTT_muon2016M_bin6
282 prop_binCROS_muon2018_bin8_Fake
283 prop_binCRTT_muon2016M_bin9
284 prop_binSR_muon2017_bin17_WrongSign
285 prop_binCRTT_electron2018_bin9
286 prop_binSR_electron2016M_bin19_WZ
287 prop_binSR_electron2017_bin4
288 prop_binCROS_electron2017_bin10
289 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
290 prop_binCROS_electron2016M_bin3
291 prop_binSR_electron2017_bin5
292 prop_binCRTT_electron2017_bin3
293 prop_binCROS_electron2017_bin8
294 prop_binSR_electron2016M_bin13_WrongSign
295 prop_binCRTT_muon2017_bin10
296 prop_binSR_electron2016M_bin3_Fake
297 prop_binCRTT_muon2017_bin5
298 prop_binCRTT_muon2017_bin13
299 prop_binCROS_electron2018_bin14
300 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

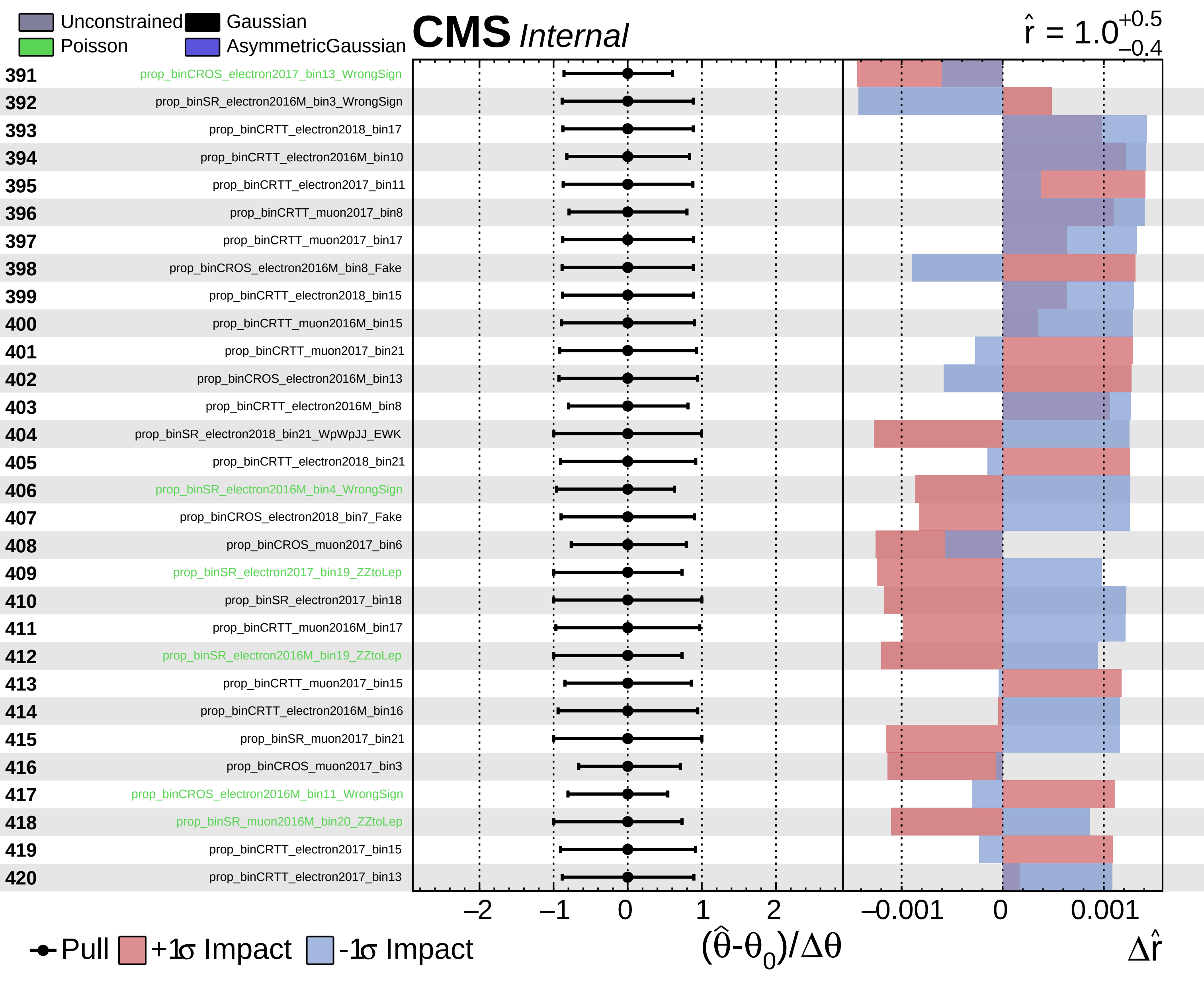

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
301 prop_binCROS_electron2018_bin15
302 prop_binSR_electron2018_bin21_TTTo2L2Nu
303 prop_binSR_electron2018_bin5
304 prop_binCRTT_electron2018_bin14
305 prop_binCRTT_muon2017_bin20
306 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
307 prop_binCRTT_electron2016M_bin3
308 prop_binSR_electron2016M_bin13_Fake
309 prop_binCRTT_muon2018_bin15
310 prop_binCROS_electron2018_bin11
311 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
312 ISR
313 prop_binCRTT_muon2016M_bin2
314 prop_binSR_muon2016M_bin5
315 prop_binSR_muon2017_bin3
316 prop_binCRTT_electron2018_bin7
317 prop_binCRTT_electron2018_bin6
318 prop_binCROS_electron2017_bin11_WrongSign
319 prop_binCRTT_electron2016M_bin2
320 mischarge_electron_2018
321 prop_binCRTT_electron2017_bin10
322 prop_binCROS_muon2016M_bin9
323 prop_binCRTT_electron2016M_bin5
324 QCDScale_VG
325 prop_binSR_electron2018_bin4
326 prop_binSR_electron2016M_bin19_TTTo2L2Nu
327 mischarge_muon_2018
328 prop_binCRTT_muon2016M_bin3
329 prop_binCRTT_electron2016M_bin12
330 prop_binCROS_muon2018_bin15
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



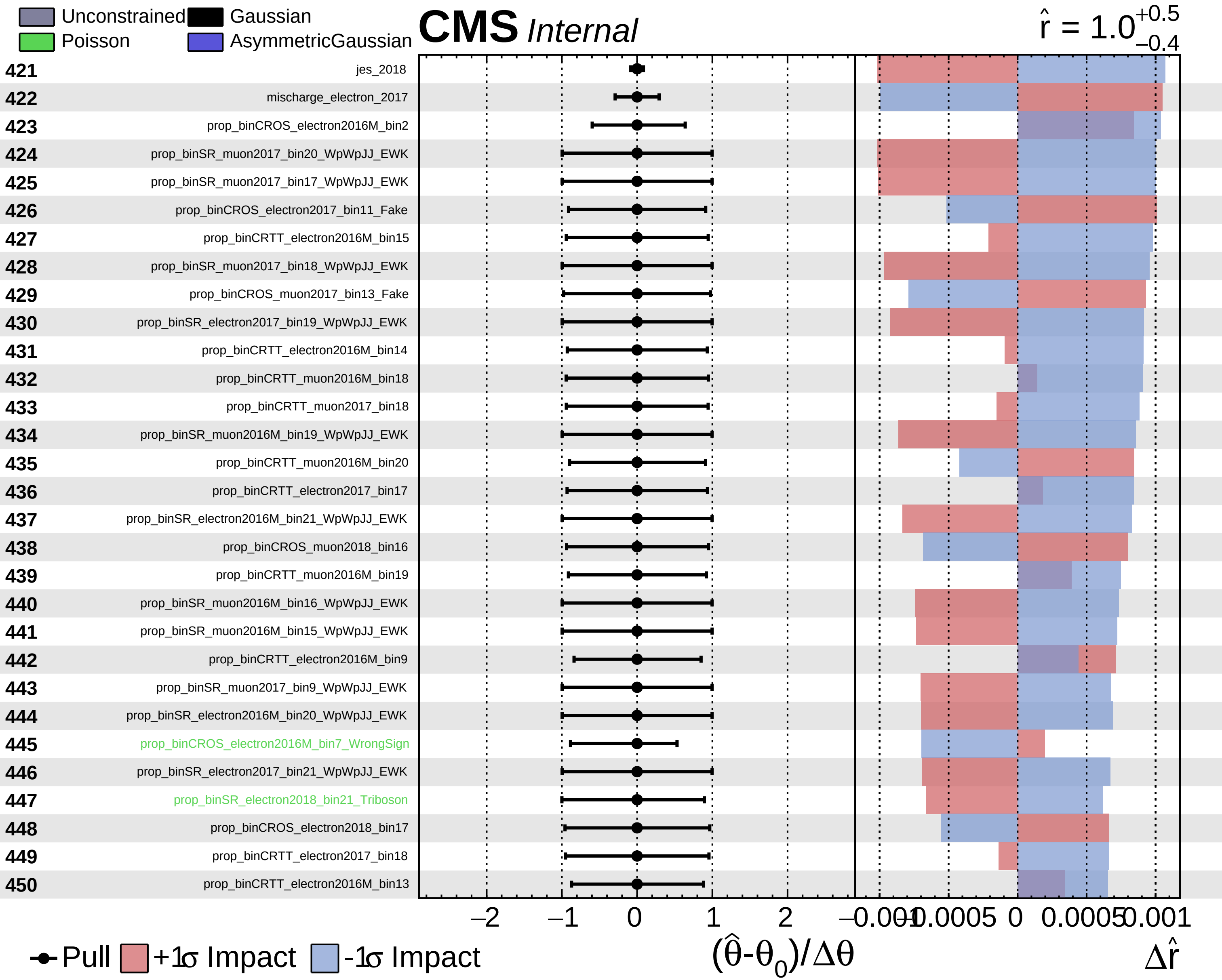
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
331 mischarge_muon_2017
332 prop_binSR_muon2016M_bin20_TTTo2L2Nu
333 prop_binCROS_muon2018_bin17_WrongSign
334 prop_binSR_electron2016M_bin13_TTTo2L2Nu
335 prop_binCRTT_muon2017_bin19
336 prop_binSR_muon2017_bin4
337 prop_binCRTT_muon2017_bin11
338 prop_binCROS_electron2018_bin10
339 prop_binSR_electron2017_bin21_TTTo2L2Nu
340 prop_binCRTT_electron2017_bin7
341 prop_binSR_electron2017_bin3
342 prop_binCRTT_muon2017_bin6
343 prop_binCRTT_electron2017_bin14
344 prop_binCRTT_muon2016M_bin13
345 prop_binSR_electron2016M_bin4_Fake
346 prop_binSR_electron2016M_bin15_Fake
347 prop_binCROS_muon2017_bin2
348 prop_binSR_muon2016M_bin4
349 puID
350 prop_binCRTT_electron2018_bin12
351 prop_binCRTT_muon2016M_bin10
352 prop_binCRTT_muon2017_bin9
353 prop_binCROS_muon2018_bin14
354 prop_binCRTT_muon2017_bin14
355 prop_binCRTT_muon2018_bin21
356 prop_binSR_muon2018_bin19_TTTo2L2Nu
357 mischarge_muon_2016M
358 prop_binSR_muon2017_bin9_Triboson
359 prop_binCROS_muon2017_bin10_Fake
360 prop_binSR_muon2017_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



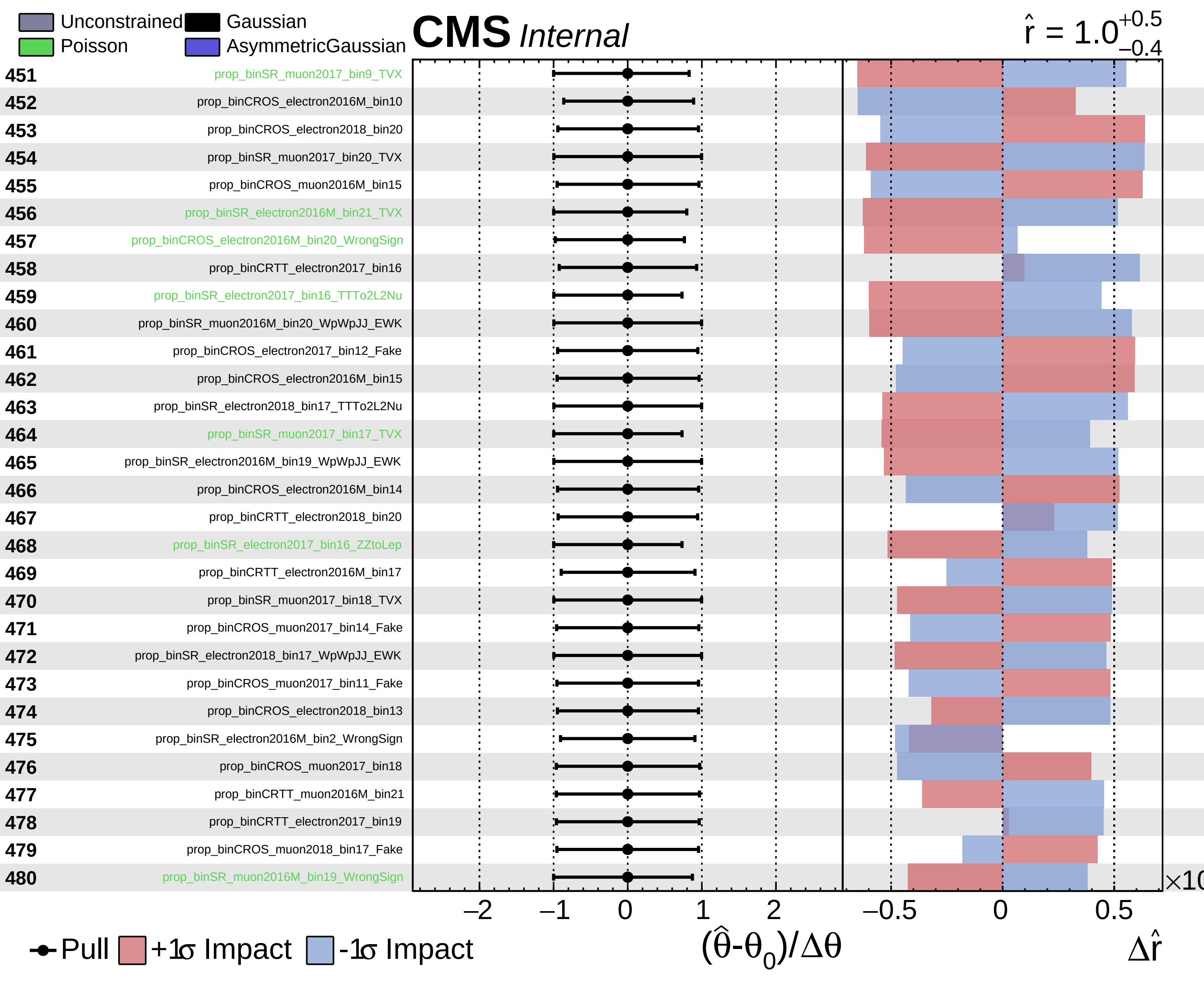
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
361 prop_binCRTT_muon2017_bin16
362 prop_binCRTT_muon2018_bin19
363 prop_binCROS_muon2016M_bin4
364 prop_binCROS_muon2017_bin17
365 prop_binSR_electron2018_bin17_WrongSign
366 prop_binCRTT_electron2017_bin12
367 prop_binSR_muon2016M_bin19_TTTo2L2Nu
368 prop_binCRTT_electron2017_bin4
369 prop_binSR_muon2018_bin19_WpWpJJ_EWK
370 prop_binCROS_electron2016M_bin6
371 prop_binSR_muon2018_bin19_ZZtoLep
372 prop_binCRTT_muon2016M_bin12
373 prop_binCRTT_muon2016M_bin14
374 prop_binCRTT_muon2016M_bin16
375 prop_binCROS_muon2018_bin18
376 prop_binCRTT_electron2017_bin6
377 prop_binCRTT_muon2016M_bin11
378 prop_binCROS_muon2017_bin16_WrongSign
379 prop_binCRTT_electron2016M_bin4
380 prop_binCRTT_muon2018_bin20
381 prop_binSR_electron2017_bin14
382 QCDScale_TTdilep
383 prop_binSR_muon2017_bin20_ZZtoLep
384 prop_binSR_electron2016M_bin2_Fake
385 prop_binCRTT_electron2016M_bin11
386 prop_binCRTT_electron2018_bin16
387 mischarge_electron_2016M
388 prop_binCRTT_electron2018_bin18
389 prop_binCROS_electron2017_bin7
390 prop_binCROS_muon2016M_bin10_Fake
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



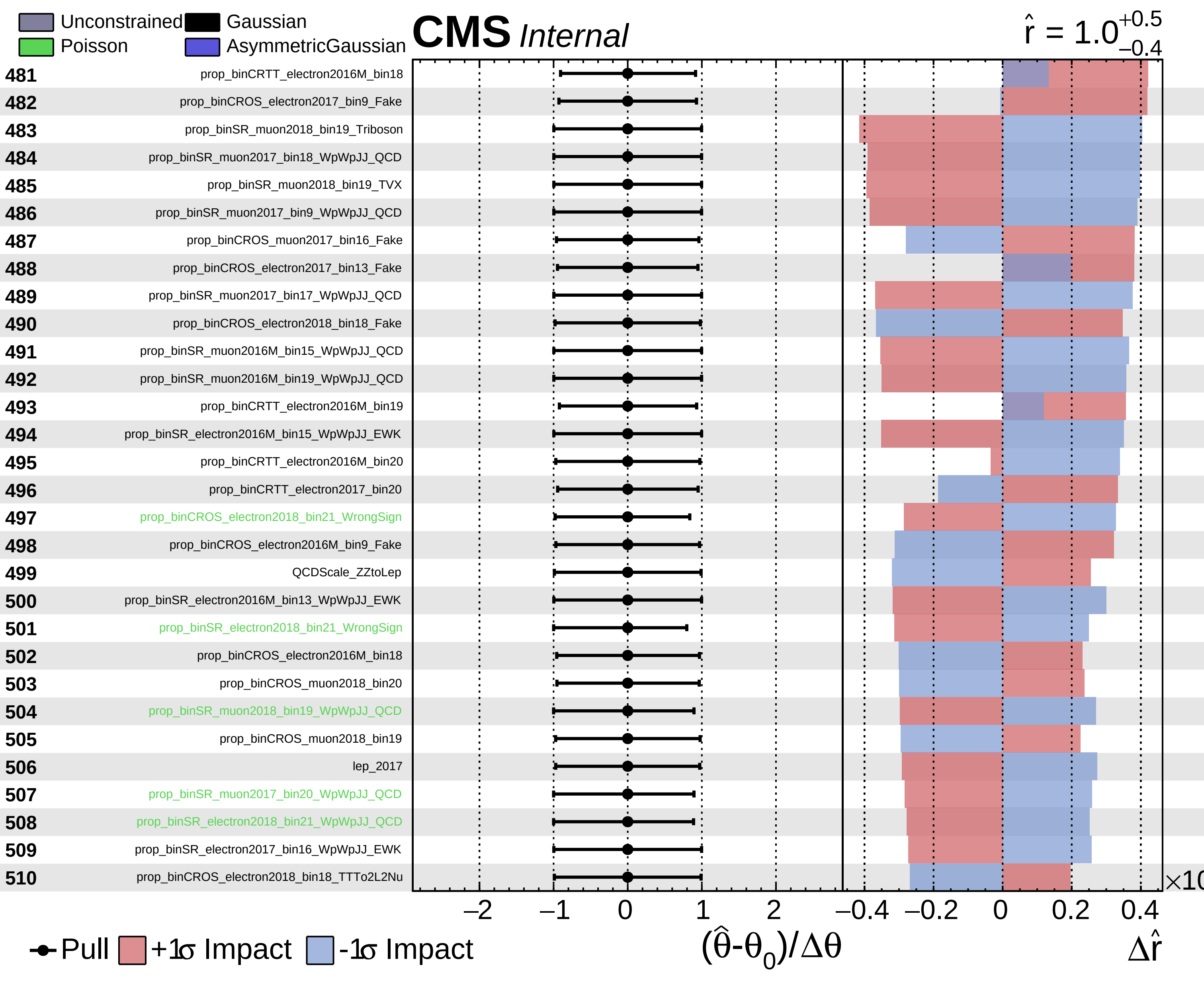
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
391 prop_binCROS_electron2017_bin13_WrongSign
392 prop_binSR_electron2016M_bin3_WrongSign
393 prop_binCRTT_electron2018_bin17
394 prop_binCRTT_electron2016M_bin10
395 prop_binCRTT_electron2017_bin11
396 prop_binCRTT_muon2017_bin8
397 prop_binCRTT_muon2017_bin17
398 prop_binCROS_electron2016M_bin8_Fake
399 prop_binCRTT_electron2018_bin15
400 prop_binCRTT_muon2016M_bin15
401 prop_binCRTT_muon2017_bin21
402 prop_binCROS_electron2016M_bin13
403 prop_binCRTT_electron2016M_bin8
404 prop_binSR_electron2018_bin21_WpWpJJ_EWK
405 prop_binCRTT_electron2018_bin21
406 prop_binSR_electron2016M_bin4_WrongSign
407 prop_binCROS_electron2018_bin7_Fake
408 prop_binCROS_muon2017_bin6
409 prop_binSR_electron2017_bin19_ZZtoLep
410 prop_binSR_electron2017_bin18
411 prop_binCRTT_muon2016M_bin17
412 prop_binSR_electron2016M_bin19_ZZtoLep
413 prop_binCRTT_muon2017_bin15
414 prop_binCRTT_electron2016M_bin16
415 prop_binSR_muon2017_bin21
416 prop_binCROS_muon2017_bin3
417 prop_binCROS_electron2016M_bin11_WrongSign
418 prop_binSR_muon2016M_bin20_ZZtoLep
419 prop_binCRTT_electron2017_bin15
420 prop_binCRTT_electron2017_bin13
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



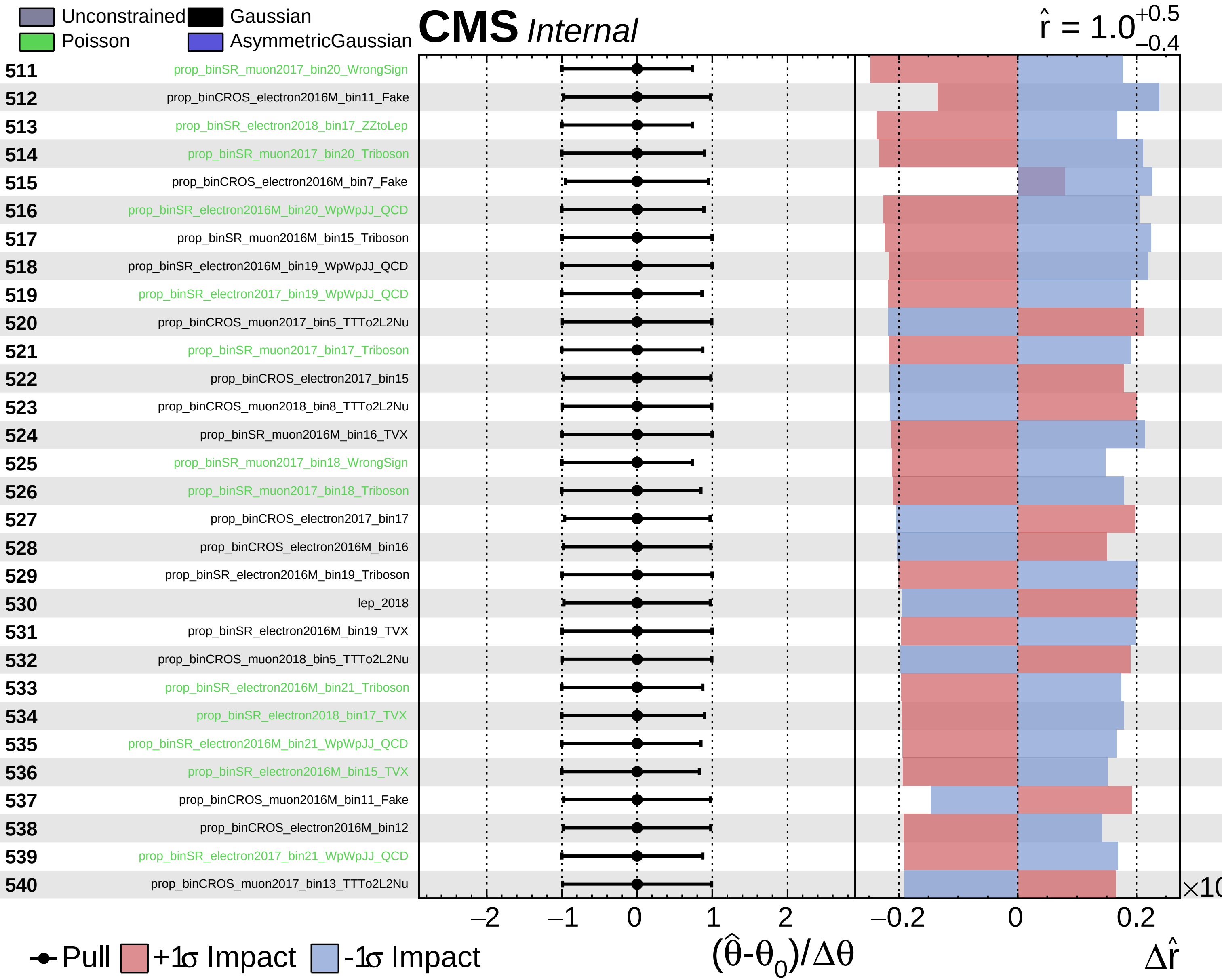
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
421 jes_2018
422 mischarge_electron_2017
423 prop_binCROS_electron2016M_bin2
424 prop_binSR_muon2017_bin20_WpWpJJ_EWK
425 prop_binSR_muon2017_bin17_WpWpJJ_EWK
426 prop_binCROS_electron2017_bin11_Fake
427 prop_binCRTT_electron2016M_bin15
428 prop_binSR_muon2017_bin18_WpWpJJ_EWK
429 prop_binCROS_muon2017_bin13_Fake
430 prop_binSR_electron2017_bin19_WpWpJJ_EWK
431 prop_binCRTT_electron2016M_bin14
432 prop_binCRTT_muon2016M_bin18
433 prop_binCRTT_muon2017_bin18
434 prop_binSR_muon2016M_bin19_WpWpJJ_EWK
435 prop_binCRTT_muon2016M_bin20
436 prop_binCRTT_electron2017_bin17
437 prop_binSR_electron2016M_bin21_WpWpJJ_EWK
438 prop_binCROS_muon2018_bin16
439 prop_binCRTT_muon2016M_bin19
440 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
441 prop_binSR_muon2016M_bin15_WpWpJJ_EWK
442 prop_binCRTT_electron2016M_bin9
443 prop_binSR_muon2017_bin9_WpWpJJ_EWK
444 prop_binSR_electron2016M_bin20_WpWpJJ_EWK
445 prop_binCROS_electron2016M_bin7_WrongSign
446 prop_binSR_electron2017_bin21_WpWpJJ_EWK
447 prop_binSR_electron2018_bin21_Triboson
448 prop_binCROS_electron2018_bin17
449 prop_binCRTT_electron2017_bin18
450 prop_binCRTT_electron2016M_bin13
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



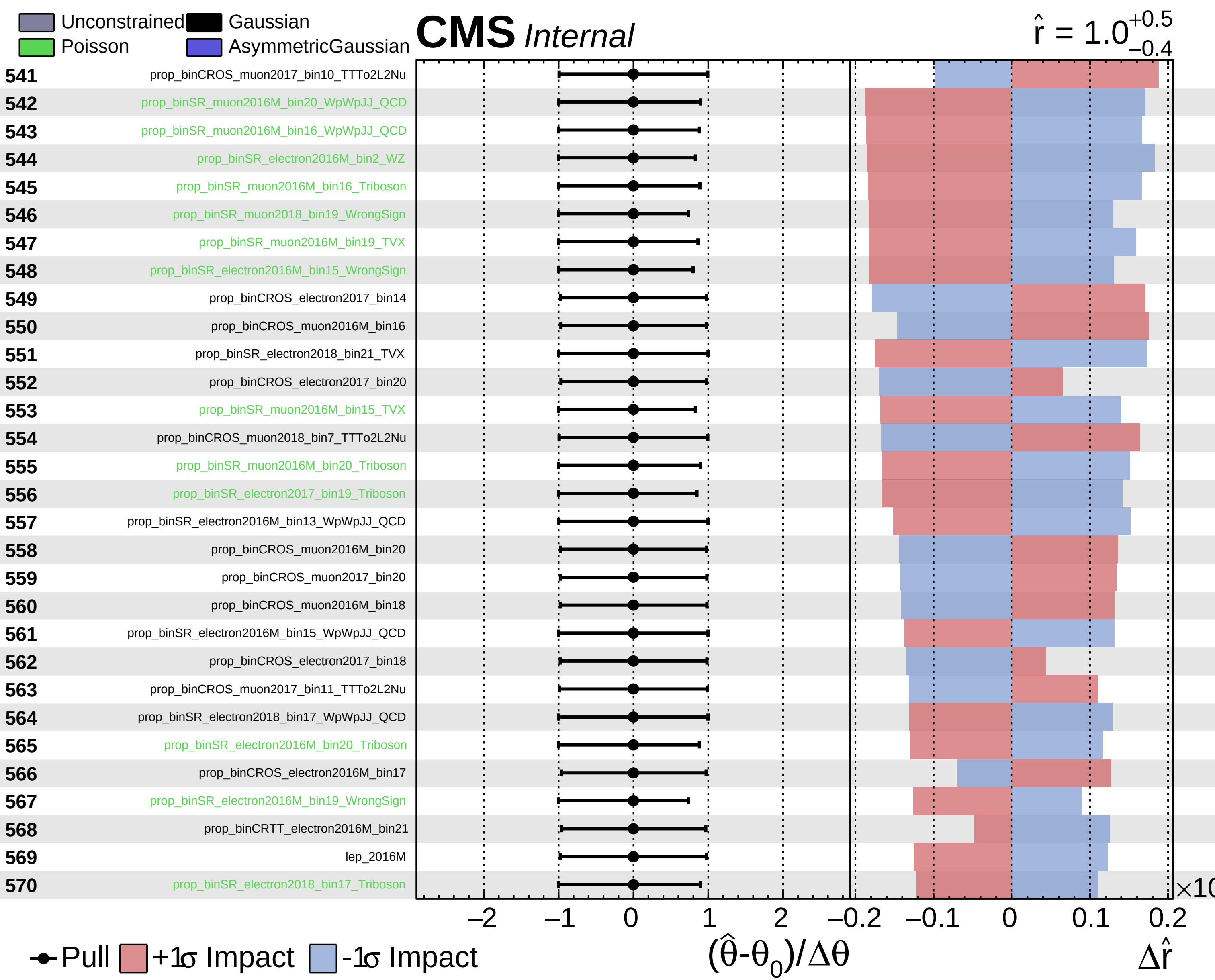
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
451 prop_binSR_muon2017_bin9_TVX
452 prop_binCROS_electron2016M_bin10
453 prop_binCROS_electron2018_bin20
454 prop_binSR_muon2017_bin20_TVX
455 prop_binCROS_muon2016M_bin15
456 prop_binSR_electron2016M_bin21_TVX
457 prop_binCROS_electron2016M_bin20_WrongSign
458 prop_binCRTT_electron2017_bin16
459 prop_binSR_electron2017_bin16_TTTo2L2Nu
460 prop_binSR_muon2016M_bin20_WpWpJJ_EWK
461 prop_binCROS_electron2017_bin12_Fake
462 prop_binCROS_electron2016M_bin15
463 prop_binSR_electron2018_bin17_TTTo2L2Nu
464 prop_binSR_muon2017_bin17_TVX
465 prop_binSR_electron2016M_bin19_WpWpJJ_EWK
466 prop_binCROS_electron2016M_bin14
467 prop_binCRTT_electron2018_bin20
468 prop_binSR_electron2017_bin16_ZZtoLep
469 prop_binCRTT_electron2016M_bin17
470 prop_binSR_muon2017_bin18_TVX
471 prop_binCROS_muon2017_bin14_Fake
472 prop_binSR_electron2018_bin17_WpWpJJ_EWK
473 prop_binCROS_muon2017_bin11_Fake
474 prop_binCROS_electron2018_bin13
475 prop_binSR_electron2016M_bin2_WrongSign
476 prop_binCROS_muon2017_bin18
477 prop_binCRTT_muon2016M_bin21
478 prop_binCRTT_electron2017_bin19
479 prop_binCROS_muon2018_bin17_Fake
480 prop_binSR_muon2016M_bin19_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

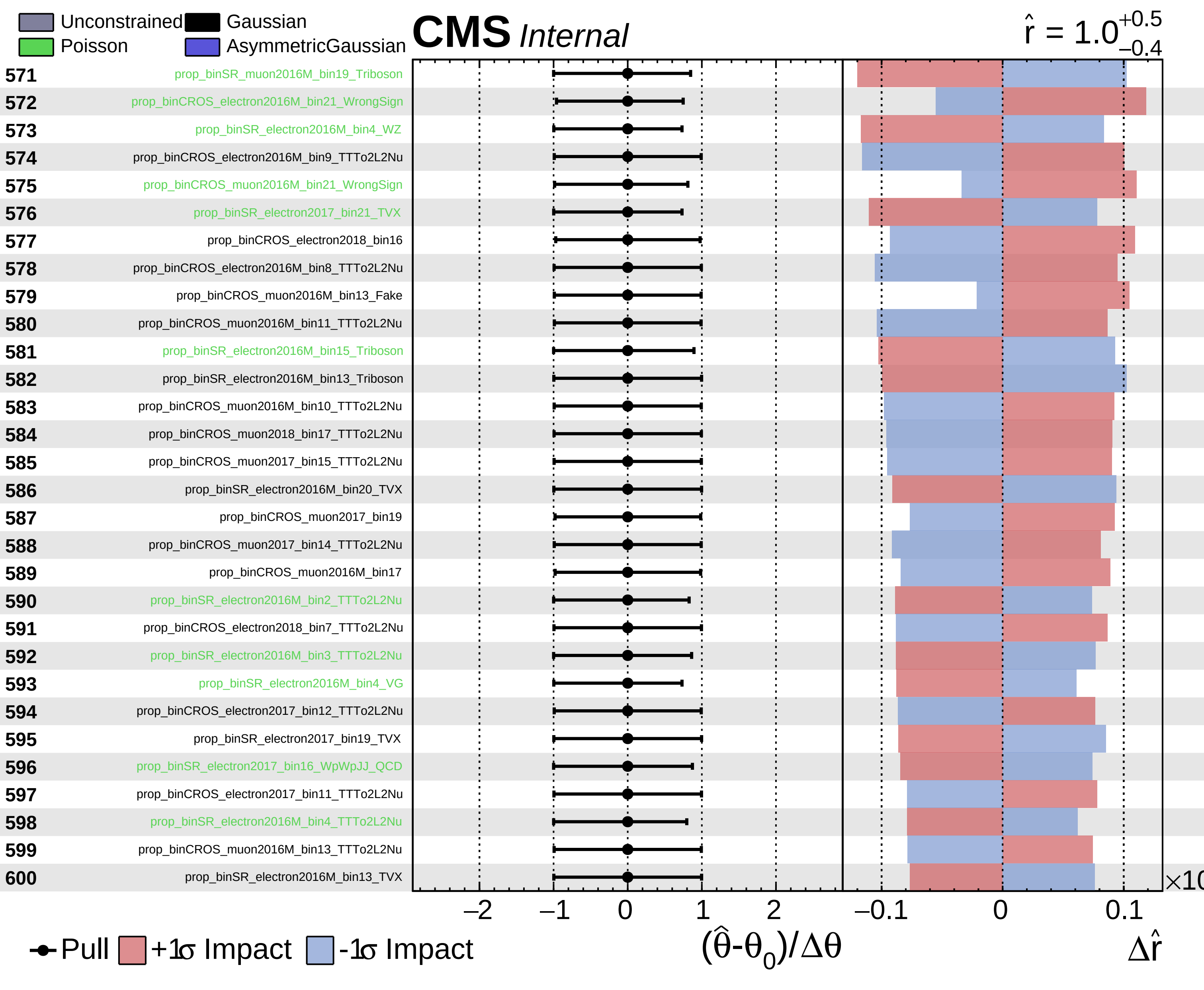


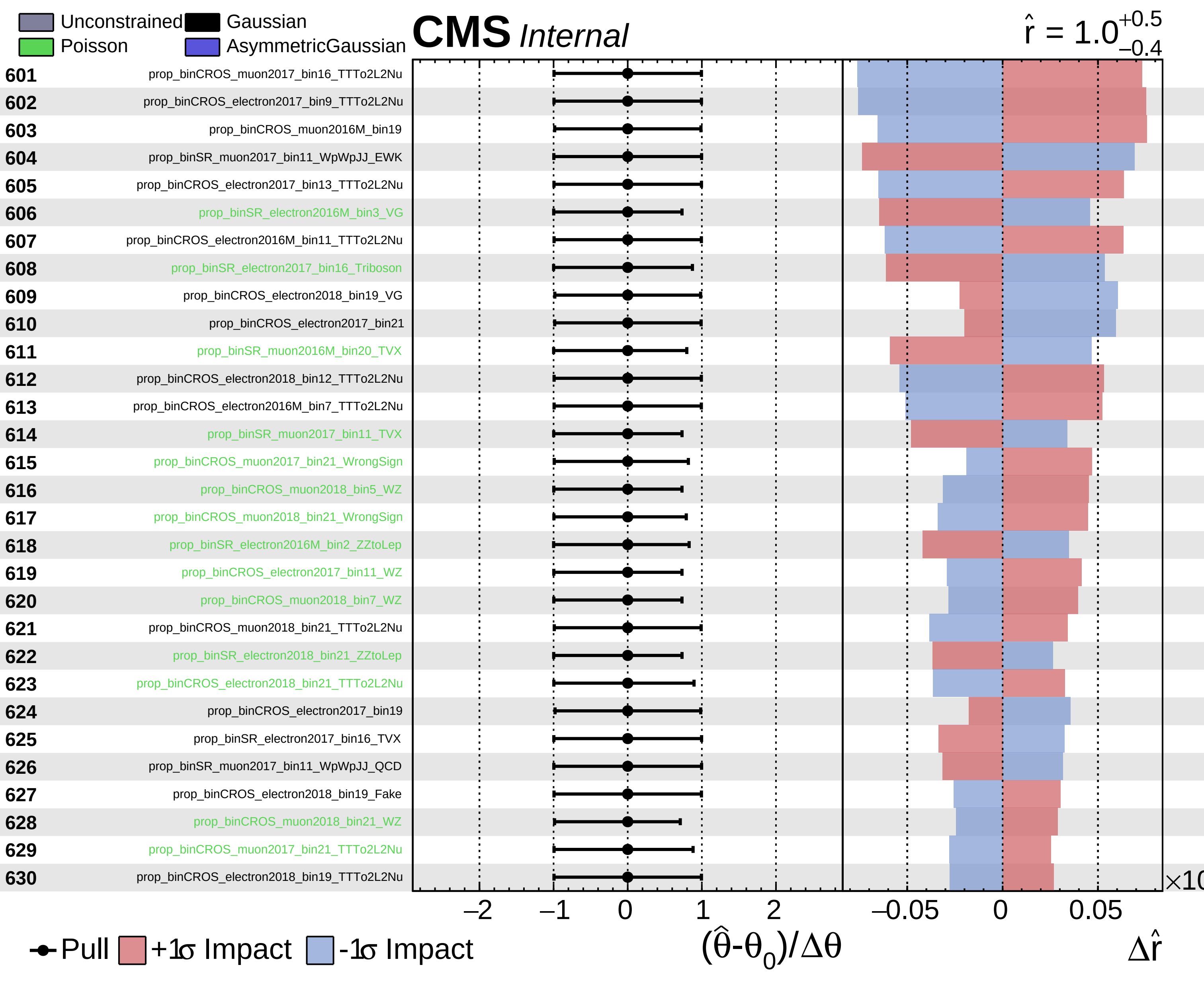
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
481 prop_binCRTT_electron2016M_bin18
482 prop_binCROS_electron2017_bin9_Fake
483 prop_binSR_muon2018_bin19_Triboson
484 prop_binSR_muon2017_bin18_WpWpJJ_QCD
485 prop_binSR_muon2018_bin19_TVX
486 prop_binSR_muon2017_bin9_WpWpJJ_QCD
487 prop_binCROS_muon2017_bin16_Fake
488 prop_binCROS_electron2017_bin13_Fake
489 prop_binSR_muon2017_bin17_WpWpJJ_QCD
490 prop_binCROS_electron2018_bin18_Fake
491 prop_binSR_muon2016M_bin15_WpWpJJ_QCD
492 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
493 prop_binCRTT_electron2016M_bin19
494 prop_binSR_electron2016M_bin15_WpWpJJ_EWK
495 prop_binCRTT_electron2016M_bin20
496 prop_binCRTT_electron2017_bin20
497 prop_binCROS_electron2018_bin21_WrongSign
498 prop_binCROS_electron2016M_bin9_Fake
499 QCDScale_ZZtoLep
500 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
501 prop_binSR_electron2018_bin21_WrongSign
502 prop_binCROS_electron2016M_bin18
503 prop_binCROS_muon2018_bin20
504 prop_binSR_muon2018_bin19_WpWpJJ_QCD
505 prop_binCROS_muon2018_bin19
506 lep_2017
507 prop_binSR_muon2017_bin20_WpWpJJ_QCD
508 prop_binSR_electron2018_bin21_WpWpJJ_QCD
509 prop_binSR_electron2017_bin16_WpWpJJ_EWK
510 prop_binCROS_electron2018_bin18_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



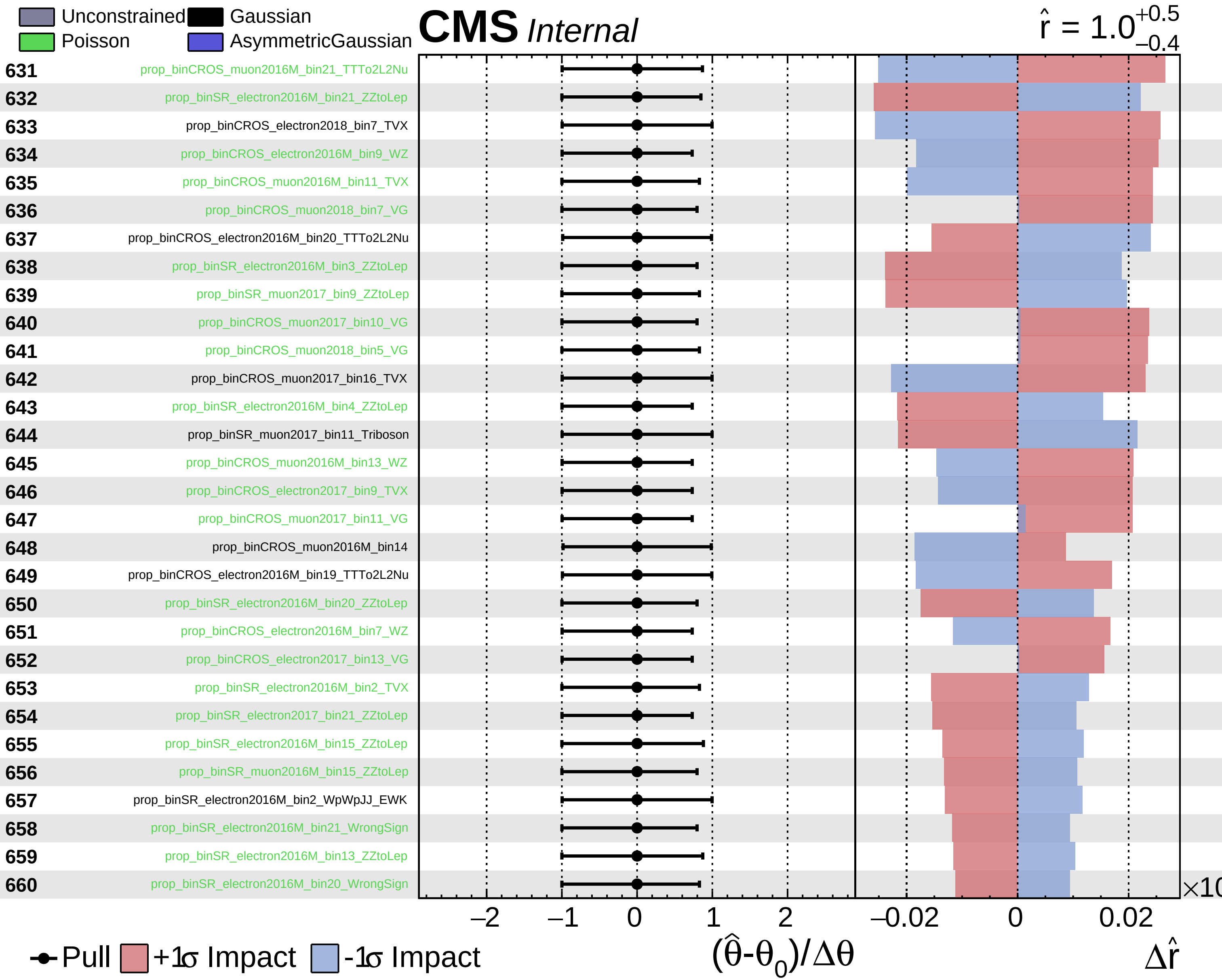
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
511 prop_binSR_muon2017_bin20_WrongSign
512 prop_binCROS_electron2016M_bin11_Fake
513 prop_binSR_electron2018_bin17_ZZtoLep
514 prop_binSR_muon2017_bin20_Triboson
515 prop_binCROS_electron2016M_bin7_Fake
516 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
517 prop_binSR_muon2016M_bin15_Triboson
518 prop_binSR_electron2016M_bin19_WpWpJJ_QCD
519 prop_binSR_electron2017_bin19_WpWpJJ_QCD
520 prop_binCROS_muon2017_bin5_TTTo2L2Nu
521 prop_binSR_muon2017_bin17_Triboson
522 prop_binCROS_electron2017_bin15
523 prop_binCROS_muon2018_bin8_TTTo2L2Nu
524 prop_binSR_muon2016M_bin16_TVX
525 prop_binSR_muon2017_bin18_WrongSign
526 prop_binSR_muon2017_bin18_Triboson
527 prop_binCROS_electron2017_bin17
528 prop_binCROS_electron2016M_bin16
529 prop_binSR_electron2016M_bin19_Triboson
530 lep_2018
531 prop_binSR_electron2016M_bin19_TVX
532 prop_binCROS_muon2018_bin5_TTTo2L2Nu
533 prop_binSR_electron2016M_bin21_Triboson
534 prop_binSR_electron2018_bin17_TVX
535 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
536 prop_binSR_electron2016M_bin15_TVX
537 prop_binCROS_muon2016M_bin11_Fake
538 prop_binCROS_electron2016M_bin12
539 prop_binSR_electron2017_bin21_WpWpJJ_QCD
540 prop_binCROS_muon2017_bin13_TTTo2L2Nu
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
541 prop_binCROS_muon2017_bin10_TTTo2L2Nu
542 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
543 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
544 prop_binSR_electron2016M_bin2_WZ
545 prop_binSR_muon2016M_bin16_Triboson
546 prop_binSR_muon2018_bin19_WrongSign
547 prop_binSR_muon2016M_bin19_TVX
548 prop_binSR_electron2016M_bin15_WrongSign
549 prop_binCROS_electron2017_bin14
550 prop_binCROS_muon2016M_bin16
551 prop_binSR_electron2018_bin21_TVX
552 prop_binCROS_electron2017_bin20
553 prop_binSR_muon2016M_bin15_TVX
554 prop_binCROS_muon2018_bin7_TTTo2L2Nu
555 prop_binSR_muon2016M_bin20_Triboson
556 prop_binSR_electron2017_bin19_Triboson
557 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
558 prop_binCROS_muon2016M_bin20
559 prop_binCROS_muon2017_bin20
560 prop_binCROS_muon2016M_bin18
561 prop_binSR_electron2016M_bin15_WpWpJJ_QCD
562 prop_binCROS_electron2017_bin18
563 prop_binCROS_muon2017_bin11_TTTo2L2Nu
564 prop_binSR_electron2018_bin17_WpWpJJ_QCD
565 prop_binSR_electron2016M_bin20_Triboson
566 prop_binCROS_electron2016M_bin17
567 prop_binSR_electron2016M_bin19_WrongSign
568 prop_binCRTT_electron2016M_bin21
569 lep_2016M
570 prop_binSR_electron2018_bin17_Triboson
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

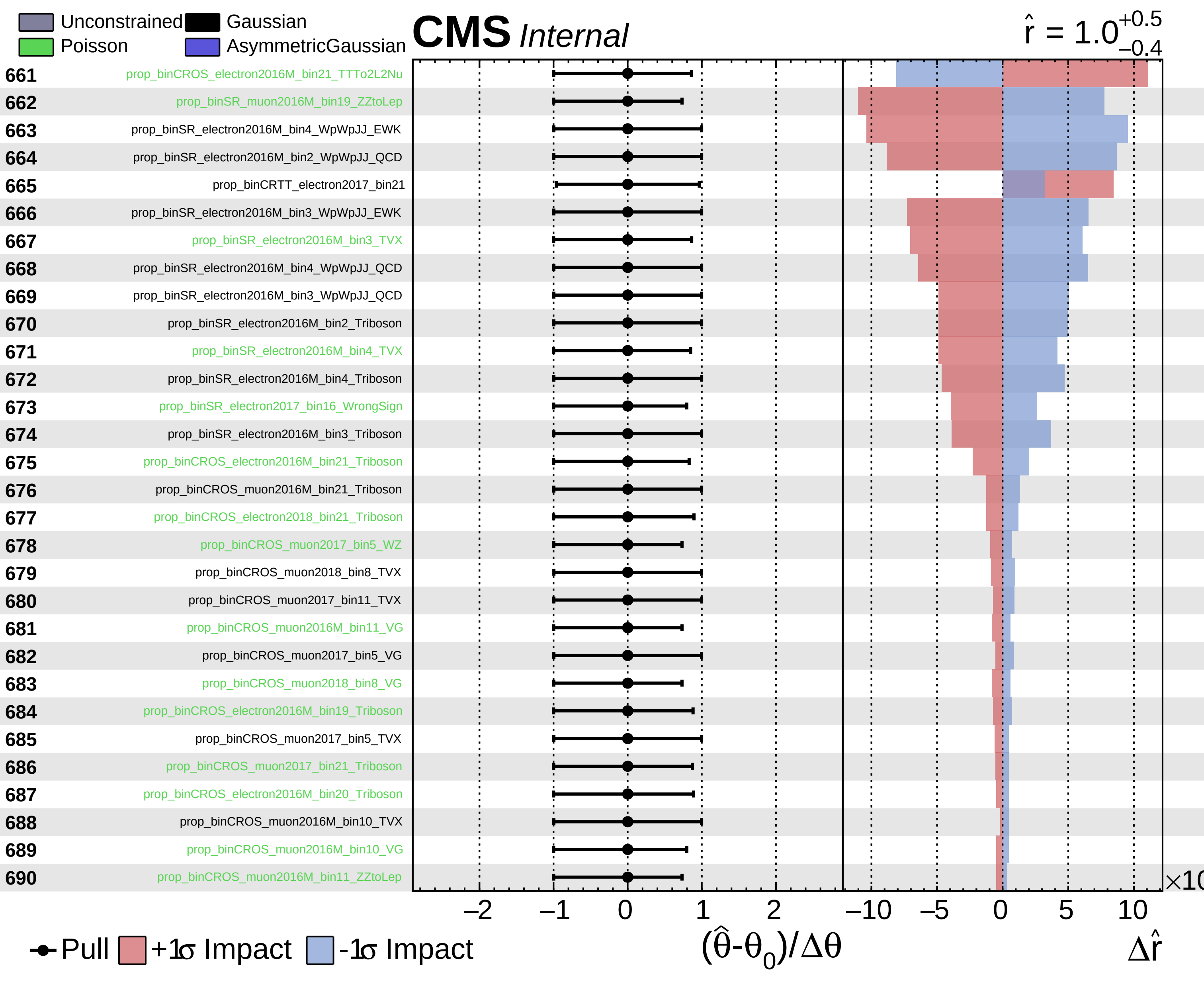

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
571 prop_binSR_muon2016M_bin19_Triboson
572 prop_binCROS_electron2016M_bin21_WrongSign
573 prop_binSR_electron2016M_bin4_WZ
574 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
575 prop_binCROS_muon2016M_bin21_WrongSign
576 prop_binSR_electron2017_bin21_TVX
577 prop_binCROS_electron2018_bin16
578 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
579 prop_binCROS_muon2016M_bin13_Fake
580 prop_binCROS_muon2016M_bin11_TTTo2L2Nu
581 prop_binSR_electron2016M_bin15_Triboson
582 prop_binSR_electron2016M_bin13_Triboson
583 prop_binCROS_muon2016M_bin10_TTTo2L2Nu
584 prop_binCROS_muon2018_bin17_TTTo2L2Nu
585 prop_binCROS_muon2017_bin15_TTTo2L2Nu
586 prop_binSR_electron2016M_bin20_TVX
587 prop_binCROS_muon2017_bin19
588 prop_binCROS_muon2017_bin14_TTTo2L2Nu
589 prop_binCROS_muon2016M_bin17
590 prop_binSR_electron2016M_bin2_TTTo2L2Nu
591 prop_binCROS_electron2018_bin7_TTTo2L2Nu
592 prop_binSR_electron2016M_bin3_TTTo2L2Nu
593 prop_binSR_electron2016M_bin4_VG
594 prop_binCROS_electron2017_bin12_TTTo2L2Nu
595 prop_binSR_electron2017_bin19_TVX
596 prop_binSR_electron2017_bin16_WpWpJJ_QCD
597 prop_binCROS_electron2017_bin11_TTTo2L2Nu
598 prop_binSR_electron2016M_bin4_TTTo2L2Nu
599 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
600 prop_binSR_electron2016M_bin13_TVX
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



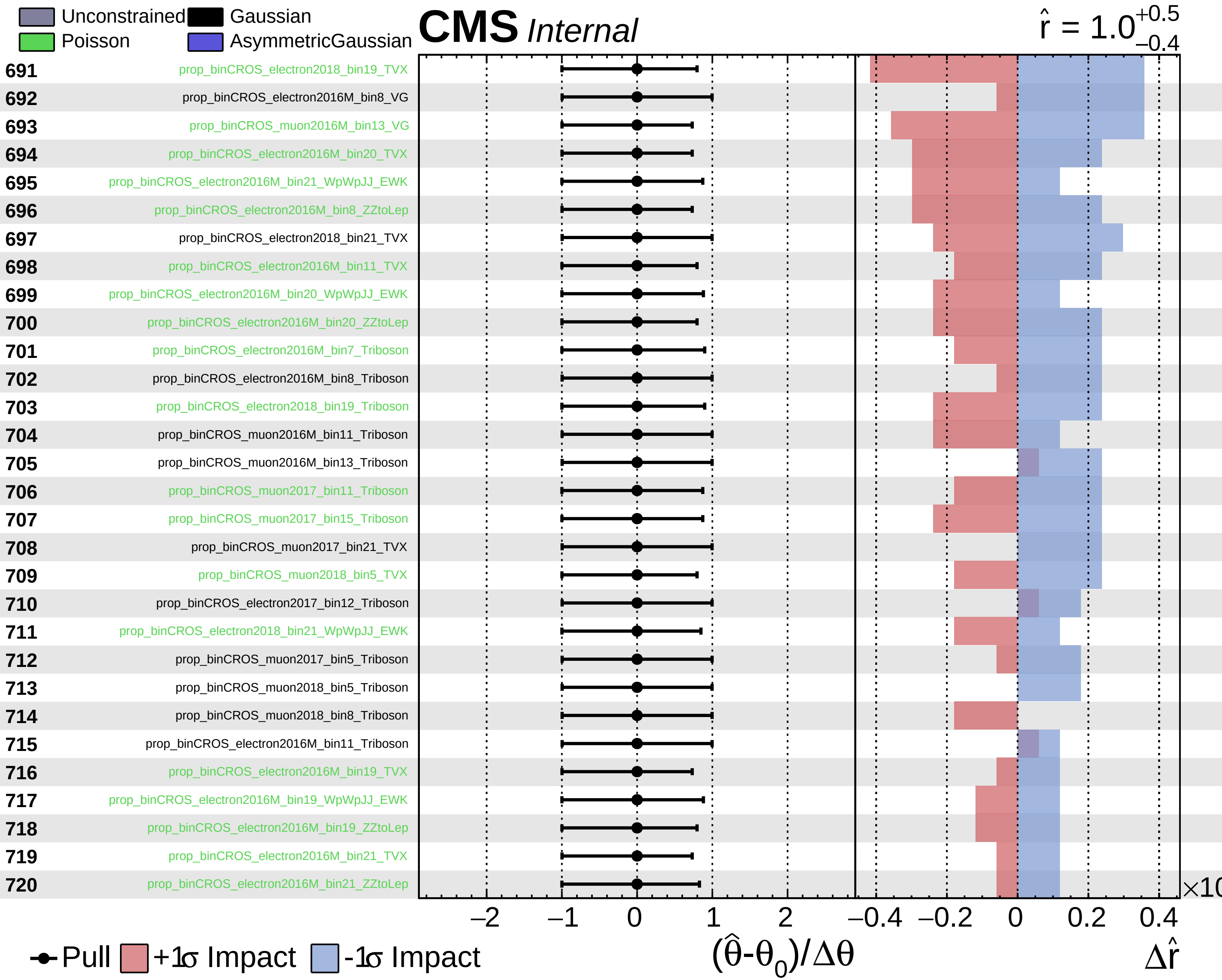
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
601 prop_binCROS_muon2017_bin16_TTTo2L2Nu
602 prop_binCROS_electron2017_bin9_TTTo2L2Nu
603 prop_binCROS_muon2016M_bin19
604 prop_binSR_muon2017_bin11_WpWpJJ_EWK
605 prop_binCROS_electron2017_bin13_TTTo2L2Nu
606 prop_binSR_electron2016M_bin3_VG
607 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
608 prop_binSR_electron2017_bin16_Triboson
609 prop_binCROS_electron2018_bin19_VG
610 prop_binCROS_electron2017_bin21
611 prop_binSR_muon2016M_bin20_TVX
612 prop_binCROS_electron2018_bin12_TTTo2L2Nu
613 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
614 prop_binSR_muon2017_bin11_TVX
615 prop_binCROS_muon2017_bin21_WrongSign
616 prop_binCROS_muon2018_bin5_WZ
617 prop_binCROS_muon2018_bin21_WrongSign
618 prop_binSR_electron2016M_bin2_ZZtoLep
619 prop_binCROS_electron2017_bin11_WZ
620 prop_binCROS_muon2018_bin7_WZ
621 prop_binCROS_muon2018_bin21_TTTo2L2Nu
622 prop_binSR_electron2018_bin21_ZZtoLep
623 prop_binCROS_electron2018_bin21_TTTo2L2Nu
624 prop_binCROS_electron2017_bin19
625 prop_binSR_electron2017_bin16_TVX
626 prop_binSR_muon2017_bin11_WpWpJJ_QCD
627 prop_binCROS_electron2018_bin19_Fake
628 prop_binCROS_muon2018_bin21_WZ
629 prop_binCROS_muon2017_bin21_TTTo2L2Nu
630 prop_binCROS_electron2018_bin19_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



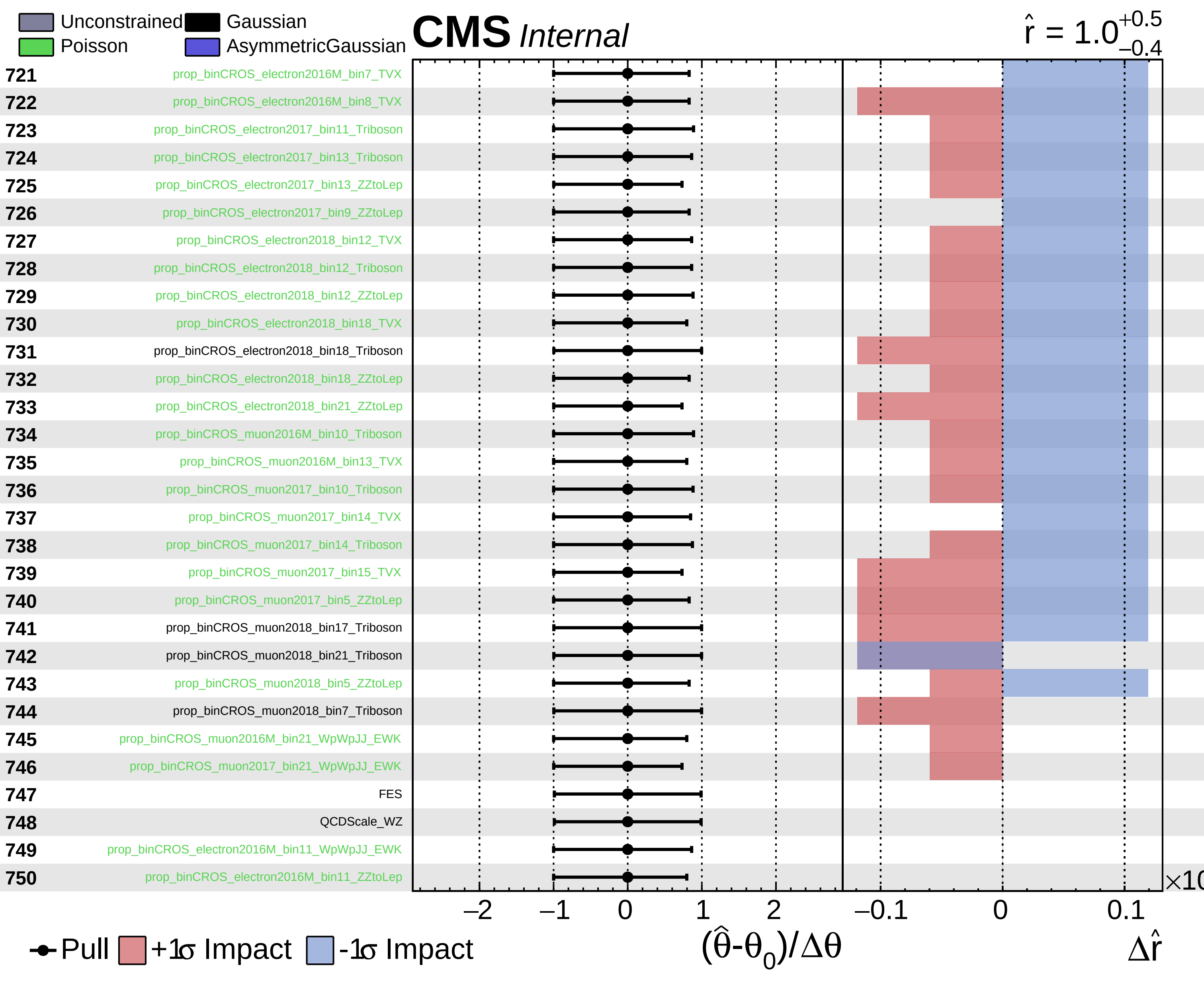
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
631 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
632 prop_binSR_electron2016M_bin21_ZZtoLep
633 prop_binCROS_electron2018_bin7_TVX
634 prop_binCROS_electron2016M_bin9_WZ
635 prop_binCROS_muon2016M_bin11_TVX
636 prop_binCROS_muon2018_bin7_VG
637 prop_binCROS_electron2016M_bin20_TTTo2L2Nu
638 prop_binSR_electron2016M_bin3_ZZtoLep
639 prop_binSR_muon2017_bin9_ZZtoLep
640 prop_binCROS_muon2017_bin10_VG
641 prop_binCROS_muon2018_bin5_VG
642 prop_binCROS_muon2017_bin16_TVX
643 prop_binSR_electron2016M_bin4_ZZtoLep
644 prop_binSR_muon2017_bin11_Triboson
645 prop_binCROS_muon2016M_bin13_WZ
646 prop_binCROS_electron2017_bin9_TVX
647 prop_binCROS_muon2017_bin11_VG
648 prop_binCROS_muon2016M_bin14
649 prop_binCROS_electron2016M_bin19_TTTo2L2Nu
650 prop_binSR_electron2016M_bin20_ZZtoLep
651 prop_binCROS_electron2016M_bin7_WZ
652 prop_binCROS_electron2017_bin13_VG
653 prop_binSR_electron2016M_bin2_TVX
654 prop_binSR_electron2017_bin21_ZZtoLep
655 prop_binSR_electron2016M_bin15_ZZtoLep
656 prop_binSR_muon2016M_bin15_ZZtoLep
657 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
658 prop_binSR_electron2016M_bin21_WrongSign
659 prop_binSR_electron2016M_bin13_ZZtoLep
660 prop_binSR_electron2016M_bin20_WrongSign
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



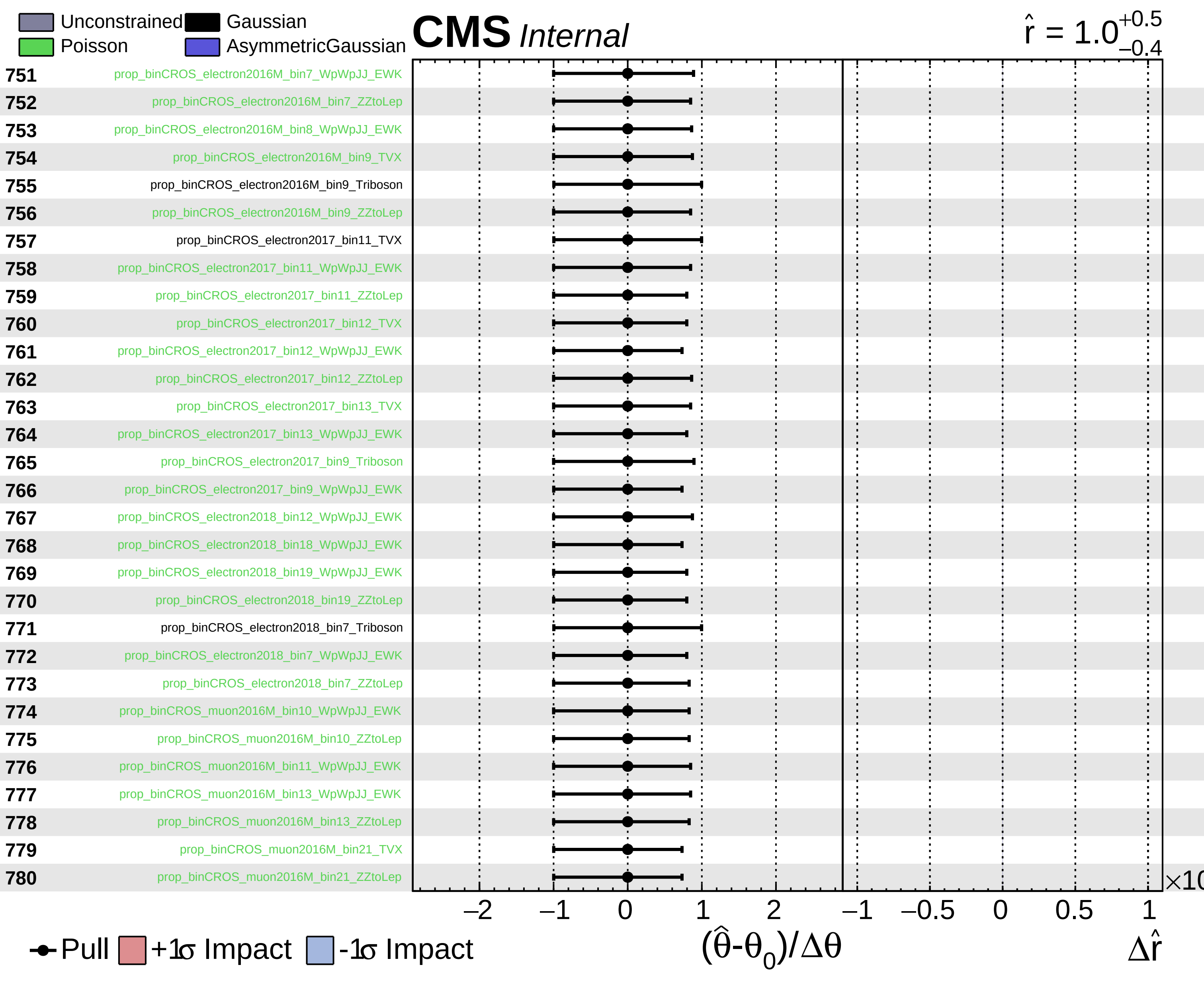
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
661 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
662 prop_binSR_muon2016M_bin19_ZZtoLep
663 prop_binSR_electron2016M_bin4_WpWpJJ_EWK
664 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
665 prop_binCRTT_electron2017_bin21
666 prop_binSR_electron2016M_bin3_WpWpJJ_EWK
667 prop_binSR_electron2016M_bin3_TVX
668 prop_binSR_electron2016M_bin4_WpWpJJ_QCD
669 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
670 prop_binSR_electron2016M_bin2_Triboson
671 prop_binSR_electron2016M_bin4_TVX
672 prop_binSR_electron2016M_bin4_Triboson
673 prop_binSR_electron2017_bin16_WrongSign
674 prop_binSR_electron2016M_bin3_Triboson
675 prop_binCROS_electron2016M_bin21_Triboson
676 prop_binCROS_muon2016M_bin21_Triboson
677 prop_binCROS_electron2018_bin21_Triboson
678 prop_binCROS_muon2017_bin5_WZ
679 prop_binCROS_muon2018_bin8_TVX
680 prop_binCROS_muon2017_bin11_TVX
681 prop_binCROS_muon2016M_bin11_VG
682 prop_binCROS_muon2017_bin5_VG
683 prop_binCROS_muon2018_bin8_VG
684 prop_binCROS_electron2016M_bin19_Triboson
685 prop_binCROS_muon2017_bin5_TVX
686 prop_binCROS_muon2017_bin21_Triboson
687 prop_binCROS_electron2016M_bin20_Triboson
688 prop_binCROS_muon2016M_bin10_TVX
689 prop_binCROS_muon2016M_bin10_VG
690 prop_binCROS_muon2016M_bin11_ZZtoLep
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

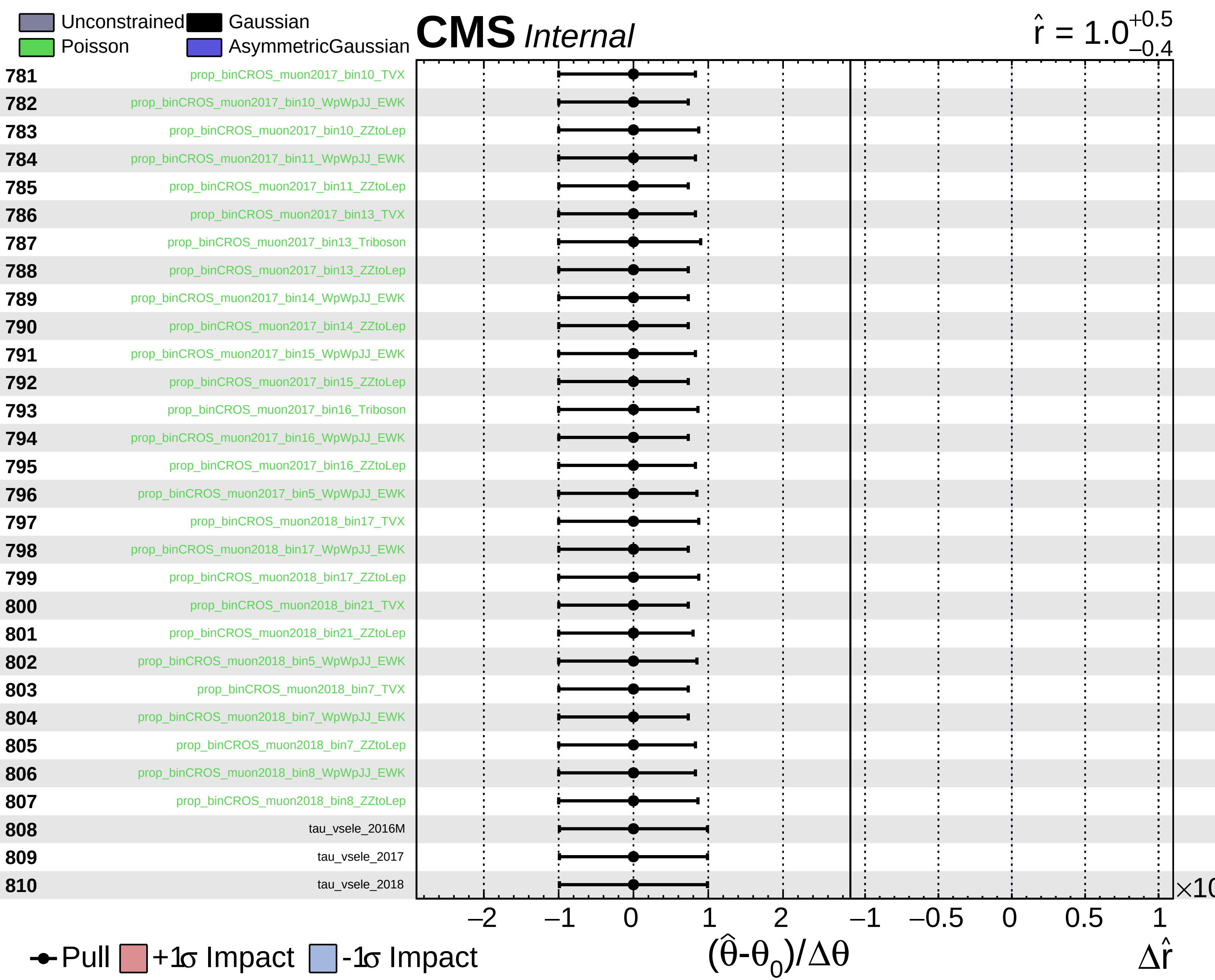


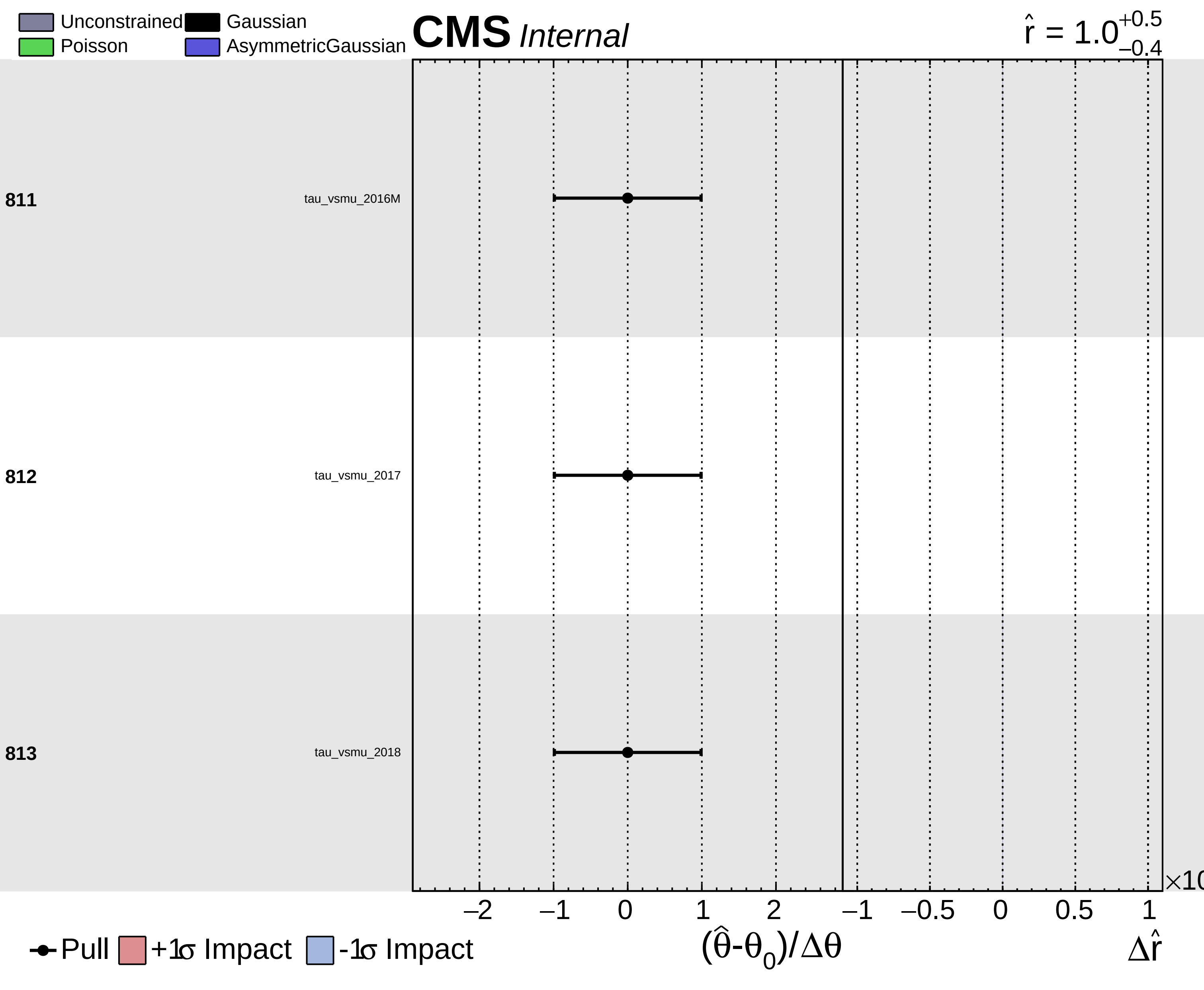
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
691 prop_binCROS_electron2018_bin19_TVX
692 prop_binCROS_electron2016M_bin8_VG
693 prop_binCROS_muon2016M_bin13_VG
694 prop_binCROS_electron2016M_bin20_TVX
695 prop_binCROS_electron2016M_bin21_WpWpJJ_EWK
696 prop_binCROS_electron2016M_bin8_ZZtoLep
697 prop_binCROS_electron2018_bin21_TVX
698 prop_binCROS_electron2016M_bin11_TVX
699 prop_binCROS_electron2016M_bin20_WpWpJJ_EWK
700 prop_binCROS_electron2016M_bin20_ZZtoLep
701 prop_binCROS_electron2016M_bin7_Triboson
702 prop_binCROS_electron2016M_bin8_Triboson
703 prop_binCROS_electron2018_bin19_Triboson
704 prop_binCROS_muon2016M_bin11_Triboson
705 prop_binCROS_muon2016M_bin13_Triboson
706 prop_binCROS_muon2017_bin11_Triboson
707 prop_binCROS_muon2017_bin15_Triboson
708 prop_binCROS_muon2017_bin21_TVX
709 prop_binCROS_muon2018_bin5_TVX
710 prop_binCROS_electron2017_bin12_Triboson
711 prop_binCROS_electron2018_bin21_WpWpJJ_EWK
712 prop_binCROS_muon2017_bin5_Triboson
713 prop_binCROS_muon2018_bin5_Triboson
714 prop_binCROS_muon2018_bin8_Triboson
715 prop_binCROS_electron2016M_bin11_Triboson
716 prop_binCROS_electron2016M_bin19_TVX
717 prop_binCROS_electron2016M_bin19_WpWpJJ_EWK
718 prop_binCROS_electron2016M_bin19_ZZtoLep
719 prop_binCROS_electron2016M_bin21_TVX
720 prop_binCROS_electron2016M_bin21_ZZtoLep
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.0^{+0.5}_{-0.4}$
721 prop_binCROS_electron2016M_bin7_TVX
722 prop_binCROS_electron2016M_bin8_TVX
723 prop_binCROS_electron2017_bin11_Triboson
724 prop_binCROS_electron2017_bin13_Triboson
725 prop_binCROS_electron2017_bin13_ZZtoLep
726 prop_binCROS_electron2017_bin9_ZZtoLep
727 prop_binCROS_electron2018_bin12_TVX
728 prop_binCROS_electron2018_bin12_Triboson
729 prop_binCROS_electron2018_bin12_ZZtoLep
730 prop_binCROS_electron2018_bin18_TVX
731 prop_binCROS_electron2018_bin18_Triboson
732 prop_binCROS_electron2018_bin18_ZZtoLep
733 prop_binCROS_electron2018_bin21_ZZtoLep
734 prop_binCROS_muon2016M_bin10_Triboson
735 prop_binCROS_muon2016M_bin13_TVX
736 prop_binCROS_muon2017_bin10_Triboson
737 prop_binCROS_muon2017_bin14_TVX
738 prop_binCROS_muon2017_bin14_Triboson
739 prop_binCROS_muon2017_bin15_TVX
740 prop_binCROS_muon2017_bin5_ZZtoLep
741 prop_binCROS_muon2018_bin17_Triboson
742 prop_binCROS_muon2018_bin21_Triboson
743 prop_binCROS_muon2018_bin5_ZZtoLep
744 prop_binCROS_muon2018_bin7_Triboson
745 prop_binCROS_muon2016M_bin21_WpWpJJ_EWK
746 prop_binCROS_muon2017_bin21_WpWpJJ_EWK
747 FES
748 QCDScale_WZ
749 prop_binCROS_electron2016M_bin11_WpWpJJ_EWK
750 prop_binCROS_electron2016M_bin11_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0+0.5 −0.4
751 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
752 prop_binCROS_electron2016M_bin7_ZZtoLep
753 prop_binCROS_electron2016M_bin8_WpWpJJ_EWK
754 prop_binCROS_electron2016M_bin9_TVX
755 prop_binCROS_electron2016M_bin9_Triboson
756 prop_binCROS_electron2016M_bin9_ZZtoLep
757 prop_binCROS_electron2017_bin11_TVX
758 prop_binCROS_electron2017_bin11_WpWpJJ_EWK
759 prop_binCROS_electron2017_bin11_ZZtoLep
760 prop_binCROS_electron2017_bin12_TVX
761 prop_binCROS_electron2017_bin12_WpWpJJ_EWK
762 prop_binCROS_electron2017_bin12_ZZtoLep
763 prop_binCROS_electron2017_bin13_TVX
764 prop_binCROS_electron2017_bin13_WpWpJJ_EWK
765 prop_binCROS_electron2017_bin9_Triboson
766 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
767 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
768 prop_binCROS_electron2018_bin18_WpWpJJ_EWK
769 prop_binCROS_electron2018_bin19_WpWpJJ_EWK
770 prop_binCROS_electron2018_bin19_ZZtoLep
771 prop_binCROS_electron2018_bin7_Triboson
772 prop_binCROS_electron2018_bin7_WpWpJJ_EWK
773 prop_binCROS_electron2018_bin7_ZZtoLep
774 prop_binCROS_muon2016M_bin10_WpWpJJ_EWK
775 prop_binCROS_muon2016M_bin10_ZZtoLep
776 prop_binCROS_muon2016M_bin11_WpWpJJ_EWK
777 prop_binCROS_muon2016M_bin13_WpWpJJ_EWK
778 prop_binCROS_muon2016M_bin13_ZZtoLep
779 prop_binCROS_muon2016M_bin21_TVX
780 prop_binCROS_muon2016M_bin21_ZZtoLep
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0 +0.5 −0.4
781 prop_binCROS_muon2017_bin10_TVX
782 prop_binCROS_muon2017_bin10_WpWpJJ_EWK
783 prop_binCROS_muon2017_bin10_ZZtoLep
784 prop_binCROS_muon2017_bin11_WpWpJJ_EWK
785 prop_binCROS_muon2017_bin11_ZZtoLep
786 prop_binCROS_muon2017_bin13_TVX
787 prop_binCROS_muon2017_bin13_Triboson
788 prop_binCROS_muon2017_bin13_ZZtoLep
789 prop_binCROS_muon2017_bin14_WpWpJJ_EWK
790 prop_binCROS_muon2017_bin14_ZZtoLep
791 prop_binCROS_muon2017_bin15_WpWpJJ_EWK
792 prop_binCROS_muon2017_bin15_ZZtoLep
793 prop_binCROS_muon2017_bin16_Triboson
794 prop_binCROS_muon2017_bin16_WpWpJJ_EWK
795 prop_binCROS_muon2017_bin16_ZZtoLep
796 prop_binCROS_muon2017_bin5_WpWpJJ_EWK
797 prop_binCROS_muon2018_bin17_TVX
798 prop_binCROS_muon2018_bin17_WpWpJJ_EWK
799 prop_binCROS_muon2018_bin17_ZZtoLep
800 prop_binCROS_muon2018_bin21_TVX
801 prop_binCROS_muon2018_bin21_ZZtoLep
802 prop_binCROS_muon2018_bin5_WpWpJJ_EWK
803 prop_binCROS_muon2018_bin7_TVX
804 prop_binCROS_muon2018_bin7_WpWpJJ_EWK
805 prop_binCROS_muon2018_bin7_ZZtoLep
806 prop_binCROS_muon2018_bin8_WpWpJJ_EWK
807 prop_binCROS_muon2018_bin8_ZZtoLep
808 tau_vsele_2016M
809 tau_vsele_2017
810 tau_vsele_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.0+0.5 −0.4
811
tau_vsmu_2016M
812
tau_vsmu_2017
813
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂